

Shoot from the HIP: Hessian Interatomic Potentials Without Derivatives

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Abstract

Fundamental tasks in computational chemistry, from transition state search to vibrational analysis, rely on molecular Hessians, which are the second derivatives of the potential energy. Yet, Hessians are computationally expensive to calculate and scale poorly with system size, with both quantum mechanical methods and neural networks. In this work, we demonstrate that Hessians can be predicted directly from a deep learning model, without relying on automatic differentiation or finite differences. We observe that one can construct SE(3)-equivariant, symmetric Hessians from irreducible representations (irrep) features up to degree $l=2$ computed during message passing in graph neural networks. This makes HIP Hessians one to two orders of magnitude faster, more accurate, more memory efficient, easier to train, and enables more favorable scaling with system size. We validate our predictions across a wide range of downstream tasks, demonstrating consistently superior performance for transition state search, accelerated geometry optimization, zero-point energy corrections, and vibrational analysis benchmarks. We open-source the HIP codebase and model weights to enable further development of the direct prediction of Hessians at <https://github.com/BurgerAndreas/hip>

1 Introduction

Over the past decades, molecular simulation has become a cornerstone for many tasks in material discovery and molecular design. Beyond energy and forces, chemists, chemical engineers, materials scientists, and physicists frequently rely on Hessians to enable critical workflows (Qu & Bowman, 2016; Huang et al., 2018). For example, *second-order geometry optimization* accelerates the determination of reliable equilibrium structures. *Transition-state searches* are needed to find reaction pathways and estimate barrier heights and transition rates. This determines whether a reaction is viable and how to tune selectivity, with high industrial relevance for the chemical and pharmaceutical industry. *Vibrational analysis* connects theory to experiment through infrared and Raman spectra

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and provide zero-point energies, which are essential for ranking isomers and estimating reaction free energies.

Despite their broad utility, Hessian calculations remain a significant computational bottleneck. For a molecule with N atoms, the Hessian is a real $3N \times 3N$ matrix, where each entry requires a mixed second derivative of the electronic energy with respect to two nuclear coordinates

$$\mathbf{H} = \nabla^2 E, \quad \mathbf{H}_{IJ}^{\alpha,\beta} = \frac{\partial^2 E}{\partial \mathbf{R}_I^\alpha \partial \mathbf{R}_J^\beta} \quad (1)$$

with $I, J = 1, \dots, N$ indexing atoms, and $\alpha, \beta \in \{x, y, z\}$ indexing the cartesian x, y and z components of the atom positions $\mathbf{R}$.

Traditional approaches rely either on analytic second derivatives or on numerical differentiation of gradients, both of which have severe limitations (Jørgensen & Simons, 1983; Handy & Schaefer, 1984; Komornicki & Fitzgerald, 1993). Analytic second derivatives are available only for a limited set of methods, such as Hartree-Fock, DFT, and MP2; whereas, for higher-level correlated wavefunction methods such as CCSD or CASPT2, they are either not yet implemented in most computational packages or computationally prohibitive. Numerical approaches sidestep analytic derivations but can become infeasible even for medium-sized molecules, requiring $O((3N)^2)$ gradient evaluations (Helgaker et al., 2000).

Most computational chemistry calculations are carried out with Kohn-Sham density functional theory (Kohn, 1965) (KS-DFT), due to its balance of computational cost and accuracy (Bursch et al., 2022). However, the $O(N^4)$ scaling of hybrid KS-DFT with system size remains expensive, which has spurred the development of machine-learning interatomic potentials (MLIPs). MLIPs promise to deliver near-DFT accuracy at a fraction of the cost by incorporating symmetries, thereby drastically reducing the need for extensive training data (Schütt et al., 2018; Thomas et al., 2018; Gasteiger et al., 2021; Schütt et al., 2021; Batatia et al., 2022; Batzner et al., 2022). While recent work demonstrates that automatic differentiation (AD) can successfully recover the Hessian (Yang et al., 2024; Yuan et al., 2024; Gönnheimer et al., 2025; Rodriguez et al., 2025; Williams et al., 2025; Cui et al., 2025), this approach faces two significant limitations. First, high accuracy of the energy and forces does not guarantee comparably accurate second derivatives; much like training on force labels is needed to achieve accurate force predictions Christensen & Von Lilienfeld (2020), explicit Hessian training labels are required for reliable AD Hessians (Zhao et al., 2025; Cui et al., 2025). Second, these AD Hessians introduce high computational costs, with an extra factor of $\sim 3N$ during training and inference (Yuan et al., 2024; Gönnheimer et al., 2025; Rodriguez et al., 2025; Williams et al., 2025; Cui et al., 2025). Rodriguez et al. Rodriguez et al. (2025) in particular showed that including the Hessian in the loss improves data efficiency and extrapolation of energy and force predictions, but is limited by approximately 25 times longer training times, which make AD Hessians prohibitive to train. MLIPs can also be used to obtain Hessians with finite differences (Gönnheimer et al., 2025; Fu et al., 2025), an approach which requires less memory but $6N$ forward passes, and also requires specialized data collection schemes to be accurate Wander et al. (2025).

In this work, we introduce **HIP: Hessian Interatomic Potentials**. HIPs predict the Hessian directly, to eliminate the need for finite differences, coupled-perturbed equation solvers, or automatic differentiation. By building on SE(3)-equivariant neural networks, we demonstrate how HIP can predict Hessians at a fraction of the computational cost of traditional methods. A key observation is that one can construct the Hessian from spherical-harmonic features of the atom nodes and the messages between atoms, satisfying equivariance and symmetry constraints by design. In practice, HIP is not only faster but also more accurate than AD, enabling a variety of critical tasks in molecular simulations.

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Our contributions are the following:

1. We present Hessian Interatomic Potentials (HIP), a method to construct SE(3)-equivariant, symmetric Hessians *directly*, without finite differences, coupled-perturbed solvers, or automatic differentiation, using an equivariant neural network backbone.
2. We show that downstream results can be improved further by using a loss function that targets the relevant Hessian subspace

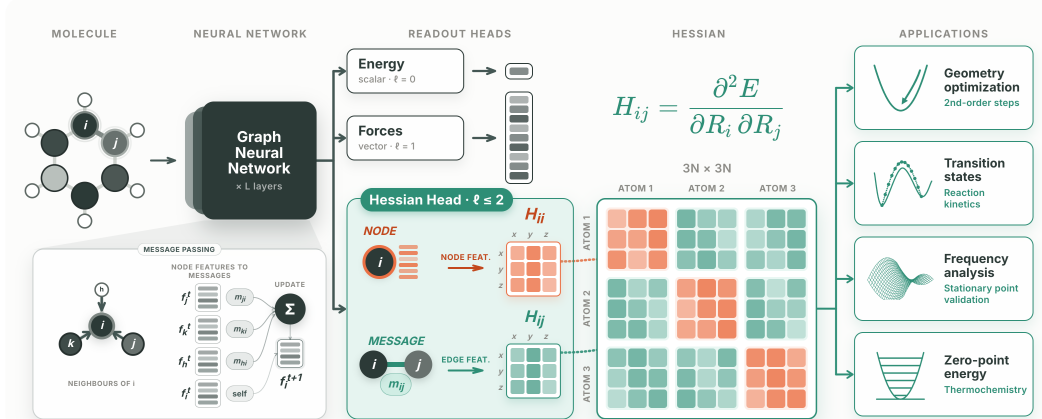


Figure 1: Regular machine learning interatomic potentials predict the energy and forces for a given molecule geometry (grey). Hessian Interatomic Potentials (HIP) expand MLIPs with a learned Hessian readout that predicts the Hessian *directly* (green and orange). In contrast, regular MLIPs rely on costly automatic differentiation (AD) to construct the Hessian matrix. HIP can use both direct force prediction and conservative forces.

3. We show that, compared to AD, the predicted Hessians are 10-100x faster, while also requiring 2-3x less peak memory on even small molecules with 5-30 atoms, with better scaling.
4. We show that the predicted Hessians are more accurate on benchmark datasets, with 2x lower Hessian MAE, 3.5x lower eigenvalue MAE, and 1.5x higher eigenvector cosine similarity, and significantly better generalization to larger systems.
5. Finally, we validate the practical utility of our predicted Hessians in several downstream applications, such as (i) zero-point energy corrections, (ii) second-order geometry optimization, (iii) transition state search, and (iv) frequency analysis for extrema classification.

2 Background

MLIPs and equivariant neural networks Early works on machine learning interatomic potentials used local atomic environment descriptors in combination with linear regression (Thompson et al., 2015; Shapeev, 2016), Gaussian processes (Bartók et al., 2010), and simple neural networks (Behler & Parrinello, 2007). SchNet (Schütt et al., 2018) was the first work to introduce *rotation-invariant graph neural networks* to predict molecular properties like energies and forces. Next-generation invariant architectures include ViSNet (Wang et al., 2022) and QuinNet (Wang et al., 2024). In parallel, *rotation-equivariant* architectures were developed, such as Cormorant (Anderson et al., 2019), DimeNet (Gasteiger et al., 2020), GemNet (Gasteiger et al., 2021), PaiNN (Schütt et al., 2021), NequIP (Batzner et al., 2022), SphereNet (Liu et al., 2022), MACE (Batatia et al., 2022) and the Equiformer family (Liao & Smidt, 2023; Liao et al., 2024). *Graph neural networks* represent a molecule as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ in 3D Euclidean space $\mathbb{R}^3$, with nodes/atoms $I \in \mathcal{V}$ at positions $\mathbf{r}_I \in \mathbb{R}^3$ and edges $(I, J) \in \mathcal{E}$ defined by a cutoff radius. At layer t , each node carries a feature $\mathbf{h}_I^{(t)}$ that is a direct sum of irreducible SO(3) representations (irreps): $\mathbf{h}_I^{(t)} = \bigoplus_{l=0}^{l_{\max}} \mathbf{h}_I^{(t,l)}$, where $\mathbf{h}_I^{(t,l)} = \{\mathbf{h}_{I,l,m}^{(t)}\}_{m=-l}^l$ has $2l + 1$ components. To update the node’s features, the model sends messages between each connected node, which are then aggregated in a permutation invariant fashion (usually the sum, or attention-weighted-sum operation) and then added to the receiving node. In SO(3)-equivariant GNNs, node features transform under a global rotation $\mathbf{Q}$ via Wigner D-matrices:

$$\mathbf{h}_{i,l,m}^{(t)}(\mathbf{Q}\{\mathbf{r}_k\}_{k=1}^N) = \sum_{m'=-l}^l \mathbf{D}_{mm'}^{(l)}(\mathbf{Q}) \mathbf{h}_{i,l,m'}^{(t)}(\{\mathbf{r}_k\}_{k=1}^N), \quad (2)$$

Translation equivariance is enforced by building messages from relative displacements $\mathbf{r}_{I,J} = \mathbf{r}_J - \mathbf{r}_I$. If we want O(3) instead of SO(3) equivariance, each order- l feature has an extra parity label that

determines the transformation under a global coordinate inversion.

Two irreps $\mathbf{p}_{l_1}$ and $\mathbf{g}_{l_2}$ interact using the Clebsch-Gordan tensor product (Thomas et al., 2018)

$$\mathbf{h}_{l_3, m_3} = \left(\mathbf{p}_{l_1, m_1} \otimes_{l_1, l_2}^{l_3} \mathbf{g}_{l_2, m_2} \right)_{m_3} = \sum_{m_1=-l_1}^{l_1} \sum_{m_2=-l_2}^{l_2} \mathbf{C}_{(l_1, m_1), (l_2, m_2)}^{(l_3, m_3)} \mathbf{p}_{l_1, m_1} \mathbf{g}_{l_2, m_2} \quad (3)$$

A message from source atom J to target atom I is then constructed as

$$\mathbf{v}_{I, J, l_3} = \mathbf{v}_{l_3}(\mathbf{h}_I, \mathbf{h}_J, \mathbf{r}_{I, J}) = \sum_{l_1, l_2} w_{l_1, l_2, l_3}(\|\mathbf{r}_{I, J}\|) \left(\mathbf{f}_{l_1}(\mathbf{h}_I, \mathbf{h}_J) \otimes_{l_1, l_2}^{l_3} \mathbf{Y}_{l_2} \left(\frac{\mathbf{r}_{I, J}}{\|\mathbf{r}_{I, J}\|} \right) \right) \quad (4)$$

where $\mathbf{Y}_{l_2}$ is the $2l_2 + 1$ dimensional vector of the spherical harmonics of degree l_2 , $w_{l_1, l_2, l_3} : \mathbb{R} \rightarrow \mathbb{R}$ is a learned weighting function, and $\mathbf{f}_{l_1}$ a simple function that takes both node features in, for example concatenation in the case of EquiformerV2. A more efficient variation of equation 4 relying on the reduction of the $SO(3)$ tensor product to the $SO(2)$ tensor products was proposed by eSCN (Passaro & Zitnick, 2023) and adopted by subsequent works, such as EquiformerV2. Finally, the messages are pooled to update the node features, which can be done, for example, via a sum or using graph attention (Liao et al., 2024).

After several layers of message passing, task-dependent readout heads map the node features to the desired targets while respecting symmetry. For example, energies E are $SO(3)$ -invariant per-graph scalars. The energy readout head therefore uses a global pooling and a reduction operation to output a $l=0$ feature. Forces $\mathbf{F}$ are $SO(3)$ equivariant per-atom vectors ($l=1$), and can be outputted either by a direct force readout head, or as the derivative of the energy $\mathbf{F} = -\nabla_{\mathbf{R}} E$. As we will see in section 3, Hessians are $SO(3)$ -equivariant per-graph matrices composed of per-atom-pair 3×3 equivariant sub-matrices.

Equivariant matrix prediction There is a rich connection between the HIP Hessian prediction in this work and other rotation-equivariant matrices of interest, like Hamiltonian prediction Kaniselman et al. (2025). For example, PhisNet Unke et al. (2021), QHNet Yu et al. (2023), HELM Kaniselman et al. (2025), and WANet Li et al. (2025) predicts the Fock matrix, and Graph2Mat Febrer et al. (2025) targets the density matrix.

2.1 Traditional methods for calculating Hessians

Hessians from DFT The gold standard for computing nuclear Hessians is to derive them analytically. In Kohn-Sham DFT, this is achieved using the Coupled Perturbed Kohn-Sham (CPKS) method (Deglmann et al., 2002), which is employed to calculate the ground truth in the dataset and in the workflows of our experiments. If available, analytical Hessians provide the most efficient ab-initio way to obtain Hessians, but they still scale as $O(N^5)$. See Appendix A.3.1 for details. A major downside of the CPKS approach to Hessians is its highly complex implementation and need for exchange-correlation-functional-specific derivations, which makes it not always available. An alternative is to use a finite-difference scheme, requiring only energies or forces. Finite differences also scale $O(N^5)$ for hybrid DFT, although with more numerical noise and a much larger computational prefactor. We provide more details in A.3.2.

Hessians from MLIPs MLIPs have gained widespread popularity for their ability to accurately approximate molecular energies and forces. The forces can either be directly predicted (Gasteiger et al., 2021; Liao et al., 2024), or calculated as the derivative of the MLIP energy using automatic differentiation. Previous work has shown that the Hessian can be calculated using AD to some success (Yang et al., 2024; Yuan et al., 2024; Gönnerheimer et al., 2025; Rodriguez et al., 2025; Williams et al., 2025; Cui et al., 2025). Although AD Hessians are much cheaper to calculate compared to DFT, they still have some downsides. First, accurately modeling the energy does not guarantee low errors on Hessians (Yuan et al., 2024), but requires dedicated training (Zhao et al., 2025; Cui et al., 2025). Second, compared to energies and forces, Hessians are expensive: Given a probing vector $\mathbf{v}$, AD only lets us calculate $\mathbf{H}\mathbf{v}$. To calculate the full $3N \times 3N$ Hessian, we need $3N$ Hessian vector products (HVPs) with the unit vectors $\mathbf{v} = \mathbf{e}_i, i = 0, \dots, 3N - 1$, each yielding one column of the Hessian. Since the energy calculation with an MLIP is usually $O(N)$, the total cost of the Hessian calculation is $O(N^2)$. While the HVPs can theoretically be parallelized, the $O(N^2)$ memory footprint requires even medium-sized systems to be computed sequentially. Rodriguez et al. (2025)

in particular showed that including the Hessian in the loss leads to better extrapolation of energy and force predictions, less data for a given accuracy, and improved stability in MD. This comes at the cost of approximately 25 longer training times, which makes AD Hessians prohibitive to train.

2.2 Molecular optimization with Hessians

Rational function optimization (RFO) Geometry optimization locates minima $\arg \min_{\mathbf{R}} E(\mathbf{R})$ of the potential energy surface (PES) starting from non-equilibrium molecular geometries. It underpins most workflows from global searches to high-throughput screening. If Hessians are available, second-order methods are favored due to their fast convergence guarantees (Nocedal & Wright, 2006; Doikov et al., 2023). RFO (Simons et al., 1983; Banerjee et al., 1985) is a commonly used second-order optimizer (it makes use of Hessian information to accelerate convergence) for molecular geometries (Hermes et al., 2022). An attractive property of RFO is that it can be used both for minimizing $E(\mathbf{R})$ as well as for finding saddle points, and it is robust to indefinite Hessians (Banerjee et al., 1985). As computing the Hessian is usually too expensive, a common practice is to maintain an approximate Hessian using the BFGS quasi-Newton scheme. We provide more details in A.4.1.

Zero-point energy (ZPE) Zero-point corrections account for the quantum mechanical vibrational energy that molecules have at absolute zero temperature. To calculate the ZPE, one relaxes a geometry to an extrema, computes the Hessian, and sums the frequencies $ZPE = \frac{\hbar}{2} \sum_i \sqrt{\tilde{\lambda}_i}$. Here, $\tilde{\lambda}_i$ are eigenvalues of the mass-weighted Hessian $\tilde{\mathbf{H}}_{AB} = \mathbf{H}_{AB}/m_A m_B$, and $\hbar$ is Planck’s reduced constant. One is usually interested in the relative ZPE between reactant and product states: $\Delta ZPE = ZPE(\mathbf{R}_R) - ZPE(\mathbf{R}_P)$. The relative ZPE is relevant for reaction thermochemistry, as ΔZPE enters as a correction to the reaction free energy ΔG and therefore the equilibrium constant, which relates forward and backward rates via detailed balance.

Transition states Transition states (TS) are first-order saddle points on the PES. They represent the maximum barrier along the minimum energy path (MEP) between two minima. Identifying transition states is crucial for understanding reaction mechanisms and mapping reaction networks. A first-order saddle point is characterized by having exactly one negative eigenvalue of the Hessian. Over the years, various computational methods have been developed to describe MEPs and locate transition states. We review methods for transition state search in A.4.2.

Frequency analysis for extrema classification To carry out frequency analysis, one must first remove the five or six redundant degrees of freedom, corresponding to the invariance of the energy under rotation and translation. This is done by mass-weighting the Hessian and performing an Eckart projection (Louck & Galbraith, 1976). Then, the projected matrix is decomposed into its eigenvalues, with all positive eigenvalues signaling a minimum, and exactly N negative eigenvalue signaling the presence of a order- N transition state.

3 HIP Hessian Prediction

Any equivariant neural network (Duval et al., 2024) with features of at least $l = 2$ can be equipped with a Hessian readout head. In this work, we pick the EquiformerV2 architecture as the backbone with four transformer layers (Liao et al., 2024). Each layer contains a layer norm, message passing with graph attention, another layer norm, and a feed-forward layer.

3.1 Hessian Symmetry Requirements

Hessians are symmetric, real-valued, Cartesian tensors. This means, under rotation of the coordinate system with rotation matrix $\mathbf{Q}$, each Hessian sub-block $\mathbf{H}_{I,J}$ transforms as

$$\mathbf{H}_{I,J} \xrightarrow{\mathbf{Q}} \mathbf{Q} \mathbf{H}_{I,J} \mathbf{Q}^\top \quad (5)$$

Any physically meaningful Hessian has to satisfy this rotation symmetry, as well as the symmetry of the upper and lower triangular $\mathbf{H} = \mathbf{H}^\top$. We elaborate on the symmetry of Hessians in A.5.

3.2 Hessian Prediction Head

Starting from T layers of message passing in the backbone, we use the atom features $\mathbf{h}_{I,m,c}^{(T)}$ and feed them into a Hessian readout module. Here, we first add another Hessian readout-specific EquiformerV2 layer to improve expressivity. We then construct the Hessian sub-blocks: We first send normal messages with equation 4, but instead of attention and summing, we treat the messages as atom-pair features:

$$\mathbf{h}_{I,J,l,m,c} = \mathbf{v}_{I,J,l,m,c} = \mathbf{v}(\mathbf{h}_I^{(T)}, \mathbf{h}_J^{(T)}, \mathbf{r}_{I,J}) \quad (6)$$

As we are reusing the message passing machinery with an interaction cutoff radius, predicting Hessians scales initially $O(N)^2$ in memory and compute within the cutoff, and then reduces to $O(N)$ scaling for larger systems. This sparsity with distance assumption aligns with the sparsity of Hessians due to locality of electron interactions that is common in quantum chemistry and can be mathematically rigorously justified (Kusmann et al., 2015). After message passing we have to transform the atom pair features $\mathbf{h}_{I,J,l,m,c}$ into the corresponding Hessian sub-blocks $\mathbf{H}_{I,J}$. First, we project the pair feature irreps down to a single $\{\tilde{\mathbf{h}}_{I,J,l,m}\}_{l \in \{0,1,2\}}$ irreps feature (1x0e + 1x1e + 1x2e in e3nn notation (Geiger & Smidt, 2022)) using a linear layer $\mathbf{W}_{l,c}$: $\tilde{\mathbf{h}}_{I,J,l,m} = \mathbf{W}_{l,c} \mathbf{h}_{I,J,l,m,c}$. We then expand $\tilde{\mathbf{h}}_{I,J,l,m}$ to an intermediate Hessian sub-block $\mathbf{H}'_{I,J}$ using the tensor product expansion (Unke et al., 2021):

$$\mathbf{H}'_{I,J,m_1,m_2} = \sum_{l,m} \mathbf{C}_{l_1=1,m_1,l_2=1,m_2}^{l,m} \tilde{\mathbf{h}}_{I,J,l,m} \quad (7)$$

where $\mathbf{C}_{l_1,m_1,l_2,m_2}^{l,m}$ are the Clebsch-Gordon coefficients ensuring that the relation in equation 5 is enforced (Unke et al., 2021). Physically, this is a simple change of basis from the coupled to the uncoupled angular momentum basis (Edmonds, 1996). $\mathbf{H}'_{I,J,m_1,m_2}$ is a 3×3 block, since m_1 and m_2 run from $-l_1 = -l_2 = -1$ to $l_1 = l_2 = 1$. As $\mathbf{C}_{l_1=1,m_1,l_2=1,m_2}^{l,m} = 0$ for all $l > 2$, $\mathbf{h}_{I,J,l,m,c}$ only needs to contain irreps up to $l = 2$ to build the 3×3 Hessian sub-blocks. Intuitively, equation 7 is the inverse of the irreducible tensor decomposition, so it assembles a spherical tensor back from its irreducible components. Conversely, this also means we need all irreps up to $l = 2$ as only including irreps up to $l = 1$ would not be sufficient to express the 3×3 tensors that decompose into $l = 2$ irreps. Finally, we symmetrize the immediate Hessian to the final prediction with $\mathbf{H} = \mathbf{H}' + \mathbf{H}'^\top$.

3.3 Loss function design

To learn the Hessians, one can use standard loss functions like mean absolute error (MAE) or mean squared error (MSE) between predicted and actual Hessian elements:

$$\mathcal{L}_{\text{MAE/MSE}} = \sum_{i,j} |\mathbf{H}_{i,j} - \mathbf{H}_{i,j}^{\text{pred}}|_{\text{MAE/MSE}} \quad (8)$$

However, we are often mainly interested in the smallest eigenvalues and the corresponding eigenvectors (Besalú & Bofill, 1998; Cerjan & Miller, 1981). For this reason, we use a loss function that emphasize the relevant subspace of Hessian. This loss function is similar to Li et al. (2025), which uses it to emphasize the subspace up to the Lowest Unoccupied Molecular Orbital (LUMO) in Fock matrix prediction. Let $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{3N}]$ be the matrix with columns made from eigenvectors of the ground truth Hessian $\mathbf{H}$, and $\mathbf{\Lambda}$ the corresponding matrix with eigenvalues on the diagonal and zeros everywhere else. Further denote $\mathbf{V}_{[:,k]}$ and $\mathbf{V}_{[:,k]}$ the matrices sliced to contain all columns up to/starting from the k^{th} one. We can then define a subspace loss by

$$\mathcal{L}_{\text{sub}} = \sum_{i,j} \left| \mathbf{V}_{[:,k]}^\top \mathbf{H}^{\text{pred}} \mathbf{V}_{[:,k]} - \mathbf{\Lambda}_{[:,k]} \right|_{i,j} \quad (9)$$

If $\mathbf{H}^{\text{pred}} = \mathbf{H}$ we have $\mathbf{V}^\top \mathbf{H}^{\text{pred}} \mathbf{V} = \mathbf{\Lambda}$, and the loss has therefore the correct minimum. Using $\mathcal{L} = \mathcal{L}_{\text{MAE/MSE}} + \alpha \mathcal{L}_{\text{sub}}$ lets us emphasize the subspace corresponding to the k lowest eigenvectors and eigenvalues. Since the Hessian always has six redundant degrees of freedom with eigenvalues close to zero, we need to set $k > 6$. In practice, we use $k = 8$. As we will show in A.7.1, predicting Hessians with HIP using any loss function significantly outperforms AD Hessians. MAE combined with the subspace loss in equation 9 further improves results in transition state search and extrema classification with frequency analysis by a small margin, which is why we use them in the experiment of the main text. For details and ablations of the choice of loss function see A.7.1.

Model	Hessian Method	Hessian Trained	$H \downarrow$ eV/Å ²	$\lambda \downarrow$ eV/Å ²	Cos $v_1 \uparrow$ unitless	$\lambda_1 \downarrow$ eV/Å ²	Time $\downarrow$ ms
AlphaNet (E-F)	AD		0.502	1.190	0.903	0.245	728.6
LEFTNet-DF (E-F)			1.650	2.247	0.505	1.362	331.4
LEFTNet-CF (E-F)			0.364	1.011	0.947	0.130	1047.9
EquiformerV2 (E-F)			2.254	4.199	0.279	1.372	564.9
AlphaNet	AD	✓	0.390	0.790	0.899	0.244	747.0
LEFTNet-DF		✓	0.200	0.172	0.937	0.136	332.2
LEFTNet-CF		✓	0.153	0.226	0.951	0.127	1051.6
EquiformerV2		✓	0.077	0.070	0.916	0.104	562.3
HIP-EquiformerV2	HIP	✓	0.030	0.063	0.982	0.031	38.5
HIP-EquiformerV2*		✓	0.020	0.041	0.982	0.031	31.4

Table 1: **Hessian prediction errors on the HORM-T1x validation set.** Denoted are the mean absolute errors (MAE), cosine similarity (Cos), and average time per forward pass on an RTX 3060 GPU. λ_1 and v_1 were obtained after Eckart-projection. Bold values highlight the best-performing model. Baseline AD models are taken from Cui et al. (2025). In HIP-EquiformerV2 we trained only the Hessian readout head while keeping the parameters of the backbone frozen, while HIP-EquiformerV2* is trained end-to-end.

4 Experiments

We evaluate our proposed approach for direct prediction of molecular Hessians across multiple tasks that test both accuracy and practical utility. Specifically, we measure prediction accuracy and quantify computational speed-ups relative to AD and finite-difference Hessian calculation. We further assess downstream performance in geometry optimization, zero-point energy, and TS searches. For TS searches, we additionally perform frequency analyses to evaluate the reliability of the predicted Hessians in distinguishing true transition states.

4.1 Dataset and training

We evaluate HIP on the Hessian dataset for Optimizing Reactive MLIP (HORM) Cui et al. (2025), comprising 1,725,362 training molecules from the Transition1x (T1x) dataset Schreiner et al. (2022), and 50,844 validation molecules. All geometries include energy, force, and Hessian labels computed from ω B97X/6-31G* density functional theory.

To isolate Hessian prediction quality from energy and force predictions, we train only the Hessian readout head while keeping the rest of EquiformerV2, pre-trained on energy, forces, and AD Hessians, frozen. This ensures that HIP-EquiformerV2 and AD-EquiformerV2 have identical energy and force predictions across all experiments. All AD-Hessian baseline models were trained on the same ground-truth DFT Hessians, so improvements are purely due to the advantage of HIP prediction over AD.

4.2 Accuracy

To quantitatively assess the accuracy and speed of our directly predicted Hessians, we calculate the average elementwise mean absolute error, average eigenvalue error, and average time per prediction. Since, for many tasks such as ZPE and TS search, the smallest eigenvalue/eigenvector pair is particularly important, we calculate the cosine similarity and eigenvalue error for these tasks separately. The results are presented in Table 1. Models trained on just energy and forces without Hessian information fail completely (EquiformerV2 (E-F) and Cui et al. (2025)). In every metric, our direct prediction Hessian approach outperforms the other models. The improvement is especially notable in prediction speed, where we are an order of magnitude faster than the models relying on automatic differentiation. Training HIP end-to-end improves the results further.

4.3 Glycine proton transfer

The aggregate errors in Table 1 summarize Hessian quality as scalar averages. To probe the curvature along a physically meaningful example, we study the intramolecular proton transfer of glycine from

the Transition1x test split Schreiner et al. (2022). The motion is described by the N–H and O–H distances of the transferring proton, q_{NH} and q_{OH} , which we transform to the reaction coordinate $\xi = q_{\text{NH}} - q_{\text{OH}}$ (see Sec. A.8.6).

We study the two-dimensional $(q_{\text{NH}}, q_{\text{OH}})$ surface around the proton-transfer transition state. At each grid point, the transferring proton is placed to match the target values of q_{NH} and q_{OH} , after which the remaining degrees of freedom are relaxed with the two distances constrained and the resulting geometry is reoptimized at the DFT level. We then map the character of the surface by counting the negative eigenvalues (Fig. 2a). HIP reproduces the DFT topology closely, while AD introduces spurious negative modes in the transition state and minima regions.

We then track the six lowest vibrational eigenvalues along the all-atom minimum energy path as a function of the reaction coordinate ξ (Fig. 2b). HIP overlaps the DFT reference across all six modes. AD Hessians capture only this first mode, but systematically underestimates the orthogonal curvatures of modes 2-6. This view matches the gap in Table 1, where HIP captures the full curvature spectrum that AD Hessians miss.

Finally, we project the Cartesian forces onto the reaction-coordinate direction $F \cdot \hat{\xi}$ along the minimum energy path (Fig. 2c). We show EquiformerV2 (EqV2) and the conservative-force model LeftNet-CF, each differentiated to obtain the AD Hessians. Both MLIPs follow the overall DFT trend (left), yet their projected force predictions oscillate significantly compared to DFT (middle). These fluctuations are amplified by AD into large Hessian errors across the reaction path (right). Together with the spurious negative modes and the underestimated orthogonal curvatures, this suggests that AD Hessian failure is driven by noise in the underlying MLIP potential energy surface.

4.4 Computational efficiency and scaling

A critical bottleneck in materials discovery is the scaling of compute with system size. Fig. 3a-b compares the inference time and memory footprint of HIP versus AD approaches. HIP exhibits the same scaling as a regular forward pass, with only a constant 25% additional time, roughly the cost of one extra layer on top of the four hidden layers. AD approaches, on the other hand, suffer from having to backpropagate through the graph for each row of the Hessian, which increases the cost to $3N$ of a forward pass. Quantitatively, HIP yields a $10\times$ to $70\times$ speedup on small molecules (5-30 atoms) and reduces peak memory usage by a factor of 2-3 compared to AD. This efficiency enables the calculation of Hessians for systems and timescales previously inaccessible to quantum mechanical levels of theory. AD calculations are also much harder to parallelize across a batch dimension; see A.6 for the explanation. By comparison, since HIP reuses the message passing machinery, HIP parallelizes just as easily as the forward energy pass. We compared the batched AD Hessian as implemented by Cui et al. (Cui et al., 2025) with our HIP-EquiformerV2. We observe at least a $78\times$ speedup in batch size compared to AD, see A.6 for more details.

4.4.1 Generalization to larger systems

To assess extrapolation beyond the training distribution, we evaluate HIP on molecules from the HORM-RGD1 dataset and 710 samples (30-100 atoms, 10 samples per number of atoms) from PubChem (Kim et al., 2025). We calculate ground truth DFT Hessians using ORCA 6.0 (Neese, 2012). Fig. 3c-d shows the error of the eigenvalues of the Hessian as a function of the number of atoms. We use the eigenvalues as a metric since their scale is independent of system size, while the MAE of Hessian elements lead to misleadingly low errors as a large share of Hessian elements go to zero for larger systems. HIP maintains strong performance across system sizes, with errors remaining relatively flat even for molecules larger than those in the training set. In contrast, the AD-based predictions show degraded performance along with the prohibitive computational costs for large molecules.

4.5 Validating downstream utility

4.5.1 Geometry optimization

We evaluate HIP Hessians in geometry optimization workflows, comparing exact second-order methods (rational function optimization, RFO) using Hessians against first-order methods and quasi-second-order RFO using BFGS. Fig. 4a shows convergence statistics across the validation set. RFO

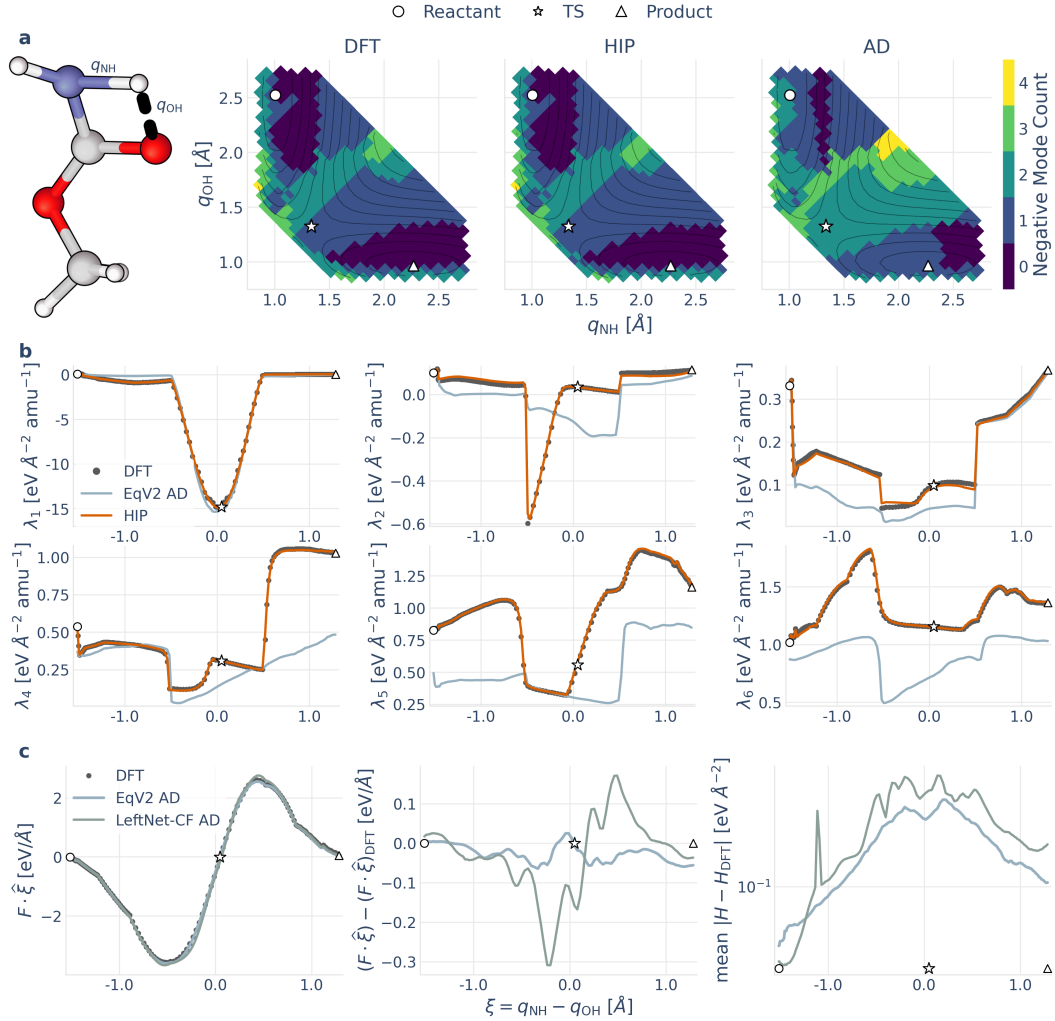


Figure 2: **Glycine proton transfer.** (a) Number of negative eigenvalues of the Eckart-projected mass-weighted Hessian over the two-dimensional collective-variable surface, for DFT, HIP, and AD. The molecule (left) shows the transferring proton and the two distances. Black lines are DFT energy contours. HIP reproduces the DFT stationary-point structure, while AD introduces spurious negative modes in the dissociating region. (b) The six lowest Hessian modes (eigenvalues of the mass-weighted Hessian) along 145 geometries of the minimum energy path, as a function of the CV $\xi = q_{\text{NH}} - q_{\text{OH}}$. DFT (black) and HIP (orange) overlap almost perfectly. AD mostly matches the first mode, but fails to reproduce modes 2 to 6. (c) Forces F used for AD, projected on the unit vector of the reaction coordinate $\hat{\xi} = \xi/|\xi|$. The predicted forces generally follow DFT (left), but oscillate significantly (middle) compared to DFT. The noise from the force predictions is amplified when taking the derivative for the AD Hessian (right).

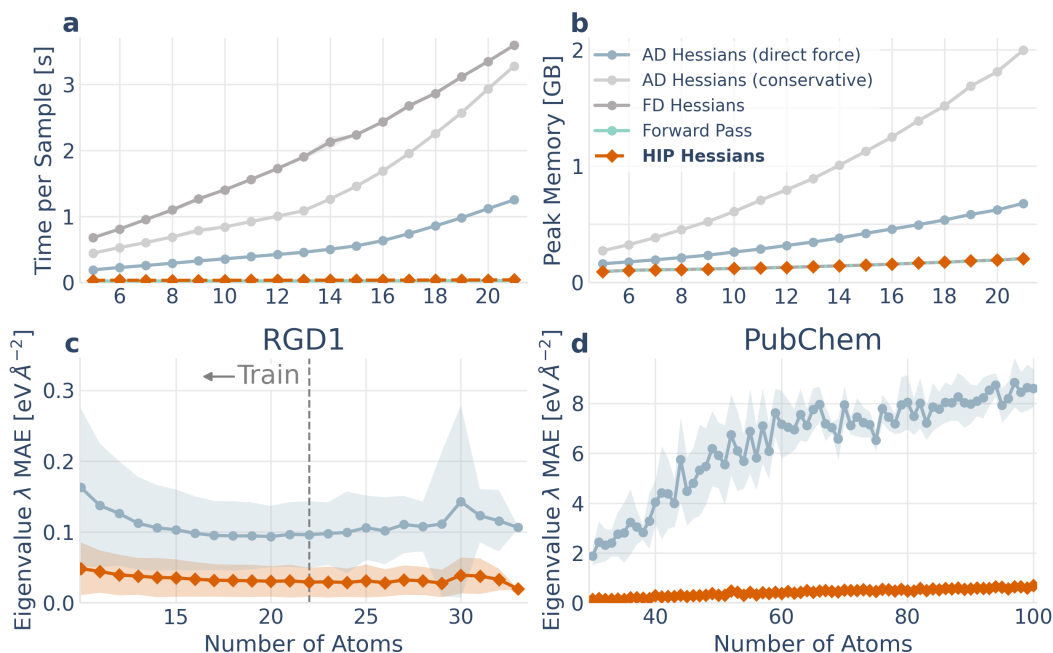


Figure 3: **Computational cost (a-b).** (a) Speed and (b) memory footprint for a single sample as a function of molecule size. Note that the same colors are used for panels (a) and (b), and that the Forward pass (shown in teal) and the HIP Hessian (this work, shown in orange) overlap. Timings were performed on an RTX 3060 GPU. **Size extrapolation (c-d).** We plot the mean and standard deviation over samples in HORM-RGD1 (c), and over ten samples per number of atoms from PubChem (d). Kim et al. (2025). For both (c) and (d), AD values are shown in blue, and HIP (this work) in orange.

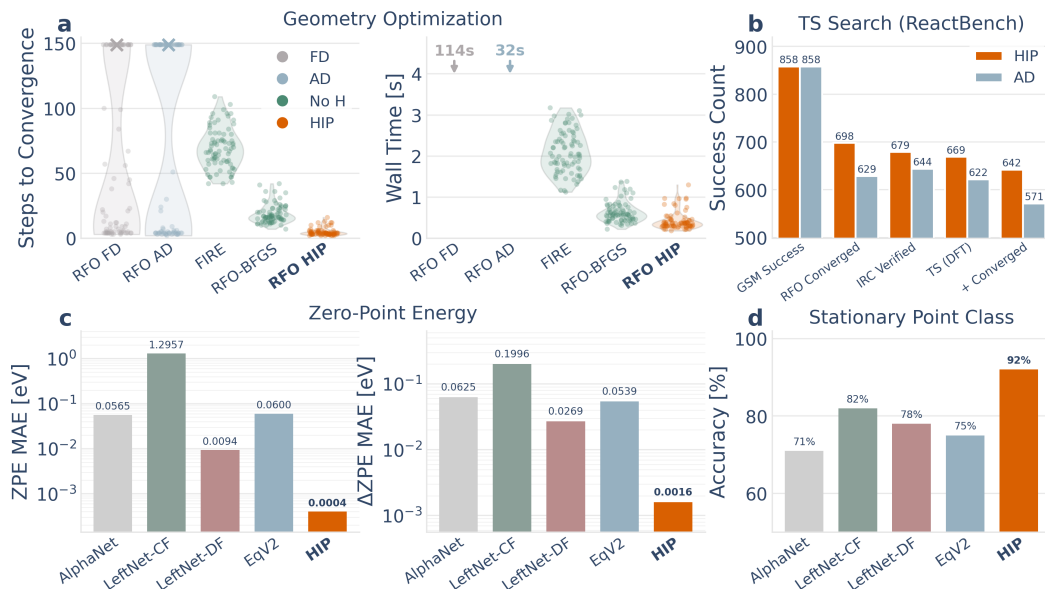


Figure 4: **Geometry optimization.** Number of steps and wall-clock time to convergence (a). For clarity, we omit the wall-clock time of RFO with AD or finite-difference Hessians, as these methods operate on a timescale one to two orders of magnitude longer than RFO with HIP-predicted Hessians. **Transition state search.** The number of successful samples for HIP Hessians and AD Hessians out of 960 initial samples on the ReactBench benchmark Zhao et al. (2025) (b). **Zero-point energy.** We report the absolute error of the ZPE of the predicted/AD Hessian relative to DFT, averaged over validation samples. We report both the ZPE at the reactant, as well as the relative Δ ZPE between reactant and product on a log scale (c). **Classifying extrema.** Panel (d) depicts the accuracy by model in characterizing stationary points via the correct number of negative Hessian eigenvalues.

with HIP Hessians converges in the fewest steps on average, whereas finite-difference and AD Hessians often fail to converge within a 150-step budget. Naïve steepest descent rarely converges, underscoring the importance of curvature information. Wall-clock timings reinforce this conclusion, as it can be observed in Fig. 4a. RFO with HIP Hessians is fastest overall, benefiting from both fewer optimization steps and efficient Hessian evaluations, while AD and finite-difference approaches are orders of magnitude slower.

4.5.2 Transition state search

Transition state search is a central challenge in computational chemistry, requiring accurate Hessian eigenvalues and eigenvectors. We evaluate HIP using the ReactBench benchmark [Zhao et al. \(2025\)](#) with additional DFT verification (A.8.3). The workflow comprises four stages: the growing string method (GSM) first approaches the TS using energies and forces; RFO optimization with Hessians then refines the TS; the intrinsic reaction coordinate (IRC) then verifies connectivity; and a final DFT Hessian confirms the saddle point. Fig. 4b shows the success rates at each stage.

GSM performs equally well for both models because it relies only on energies and forces, which are identical. The differences emerge in Hessian-dependent stages. HIP achieves higher RFO convergence, better IRC verification, and most importantly, more DFT-verified transition states. When requiring both DFT verification and tight convergence, HIP improves 12% over AD. These results demonstrate that HIP’s superior Hessian quality translates to more reliable TS identification, which is crucial for estimating reaction barriers and rates.

4.5.3 Zero-point energy

Zero-point energy is a critical thermochemical property derived from vibrational frequencies, which depend directly on Hessian eigenvalues. We evaluate both absolute ZPE at the reactant geometry and relative Δ ZPE between reactants and products. Fig. 4c reveals dramatic improvements with HIP. For absolute ZPE, HIP achieves an order of magnitude better accuracy than LeftNet-DF and two orders of magnitude better than AD-EquiformerV2. For relative Δ ZPE, HIP similarly reduces the error by 97.03% compared to AD-EquiformerV2. These results demonstrate that HIP’s improved eigenvalue predictions translate directly to better thermochemical properties, where small errors in vibrational frequencies can lead to large errors in free energy.

4.5.4 Frequency analysis for extrema classification

Characterizing stationary points on potential energy surfaces requires determining the number of negative Hessian eigenvalues: zero for minima, one for first-order transition states, and two or more for higher-index saddle points [Maronsson et al. \(2012\)](#). We test this capability on 1000 geometries from the HORM-T1x validation, comprising 44% first-order TSs, 37% minima, and 19% higher-order saddle points, as determined from DFT Hessians. Fig. 4d shows classification accuracy based on counting negative eigenvalues. HIP achieves 92% accuracy, 10 percentage points better than the next-best AD model and 17 percentage points better than AD-EquiformerV2. This substantial improvement stems from HIP’s more accurate eigenvalue predictions, particularly for the small eigenvalues that determine the character of the stationary point. Accurate extrema classification is essential for automated reaction pathway discovery and for validating that optimizations have reached the intended type of stationary point.

4.6 Synergistic improvements from Hessian training

Thus far, we trained only the Hessian readout while freezing the backbone, isolating Hessian quality from energy and force predictions. We now investigate joint end-to-end training of the full model, including Hessian supervision. We train both the standard EquiformerV2 (energy and forces only) and the HIP-EquiformerV2 (energy, forces, and Hessians) for equal wall-clock time (3 days) across various dataset sizes. Fig. 5a-c reveals that Hessian supervision improves not only Hessian predictions but also energy and force accuracy across all data regimes.

On 10,000 validation structures (Fig. 5d-f), force and Hessian errors are positively correlated for Hessian-trained models, but nearly independent for an energy-force-only baseline, which can achieve good force accuracy while Hessian MAE remains an order of magnitude higher. HIP occupies a tighter, lower-error region than AD, indicating a more self-consistent potential energy surface.

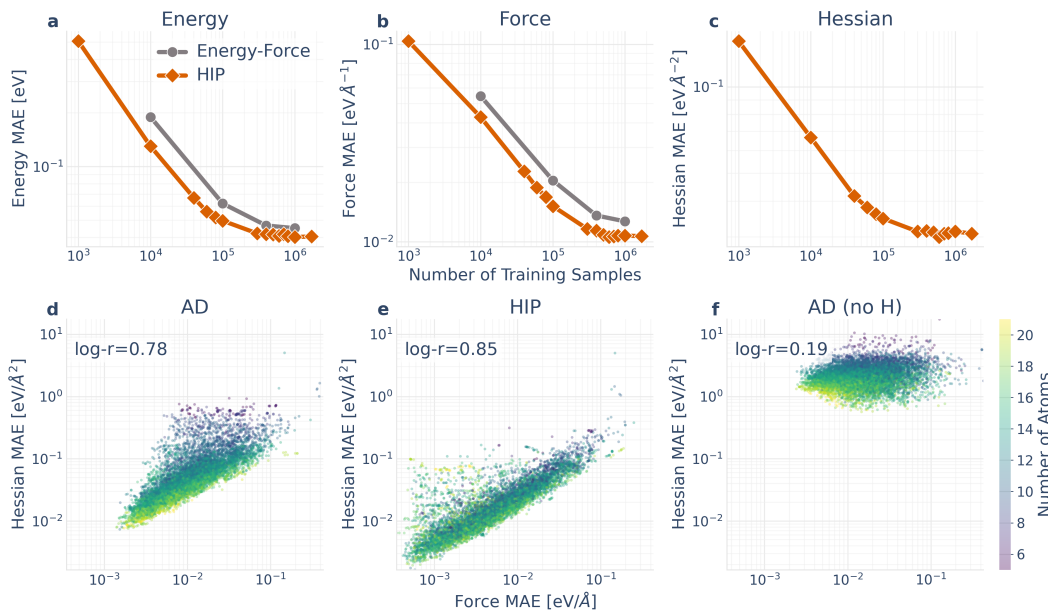


Figure 5: **Accuracy scaling with dataset size.** (a) Energy, (b) Force, and (c) Hessian MAE on the HORM-T1x validation set as a function of the number of training samples randomly drawn from the HORM-T1x training set. The EquiformerV2 trained on energy and forces is shown in dark gray, whereas the same model trained as HIP end-to-end on Hessian data (this work) is shown in orange. **Correlation between Hessian and force accuracy.** (d-f) show 10,000 data points of the HORM-T1x validation set of (d) EquiformerV2 with the HORM Hessian training (e) our HIP-EquiformerV2* (f) EquiformerV2 trained on only energies and forces.

This synergistic effect suggests that Hessian information provides a richer training signal, helping the model learn a better potential energy surface representation.

Non-Conservative Hessians Because HIP predicts Hessians directly rather than as exact second derivatives of a scalar energy, the predicted Hessians are not guaranteed to be globally integrable. In contrast to long-time molecular dynamics simulations, where non-integrability errors of the forces accumulate unbounded over time, all common workflows involving Hessians are insensitive to non-integrability errors. This is already evidenced by the success of the Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm, a classical tool to approximate Hessians that does not satisfy any notion of integrability. A more mathematical discussion about this topic is provided in the Supplementary Information A.5.1.

Limitations Direct Hessian regression requires ground-truth data from DFT, similar to the energy and force labels used for regular MLIPs. The experiments in this work focus on small organic molecules, and future work should explore applying Hessian prediction to other chemical systems, such as materials, larger complexes, or even proteins. To isolate the Hessian from the force and energy predictions, we trained only the Hessian regression head using a fixed backbone. For optimal results on all benchmarks, energy, forces, and Hessians should be trained end-to-end.

5 Conclusion

In this work, we have presented Hessian Interatomic Potentials (HIP), a novel method for directly predicting molecular Hessians using SE(3)-equivariant neural networks. Our approach eliminates the need for traditional computationally expensive methods, such as finite differences or automatic differentiation, offering a significant improvement in computational time, memory usage, and accuracy. Through extensive validation across a range of molecular tasks, we have shown that our predicted Hessians are highly effective for practical applications in computational chemistry.

Looking forward, the architectural flexibility of HIP, which can be integrated into any message-passing base model built on spherical harmonics, opens significant opportunities beyond isolated molecules. By making accurate Hessians computationally inexpensive, this framework enables high-throughput automated reaction pathway discovery, materials design and drug discovery at previously inaccessible scales. A particularly promising direction is the straightforward adaptation of HIP to systems with periodic boundary conditions. Because HIP constructs the Hessian locally from atom-pair interactions within a cutoff radius, it inherently supports periodic boundary conditions via a standard periodic neighbour list. HIP could also be extended to the direction prediction of higher-rank objects like the elastic tensor. The rank-4 nature of the elastic tensor, the second derivatives of the energy with respect to macroscopic strain, could be learned by additionally building global $l = 4$ features. Due to the versatility of the framework, any base model using spherical harmonics and message passing can be made HIP.

Data and code availability

The training and validation data is available at www.kaggle.com/datasets/yunhonghan/hessian-dataset-for-optimizing-reactive-mlipform/data. Checkpoints and additional DFT calculations for validation are available under huggingface.co/andreasburger/hip. The training and validation code is available under github.com/BurgerAndreas/hip. All requests should be addressed to Andreas Burger via me@andreas-burger.com.

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A Appendix

A.1 Extended background

A.2 MLIPs and equivariant neural networks

Early works on machine learning interatomic potentials used local atomic environment descriptors in combination with linear regression (Thompson et al., 2015; Shapeev, 2016), Gaussian processes (Bartók et al., 2010), and simple neural networks (Behler & Parrinello, 2007). SchNet (Schütt et al., 2018) was the first work to introduce *rotation-invariant graph neural networks* for predicting molecular properties such as energies and forces. Next-generation invariant architectures include ViSNet (Wang et al., 2022) and QuinNet (Wang et al., 2024). In parallel, *rotation-equivariant* architectures were developed, such as Cormorant (Anderson et al., 2019), DimeNet (Gasteiger et al., 2020), GemNet (Gasteiger et al., 2021), PaiNN (Schütt et al., 2021), NequIP (Batzner et al., 2022), SphereNet (Liu et al., 2022), MACE (Batatia et al., 2022) and the Equiformer family (Liao & Smidt, 2023; Liao et al., 2024; Burger et al., 2025).

After several layers of message passing, task-dependent readout heads map the node features to the desired targets while respecting symmetry. For example, energies E are $SO(3)$ -invariant per-graph scalars. The energy readout head therefore uses a global pooling and a reduction operation to output a $l=0$ feature. Forces $\mathbf{F}$ are $SO(3)$ equivariant per-atom vectors ($l = 1$), and can be outputted either by a direct force readout head, or as the derivative of the energy $\mathbf{F} = -\nabla_{\mathbf{R}} E$. Hessians are $SO(3)$ -equivariant per-graph matrices composed of per-atom-pair 3×3 equivariant sub-matrices. There is a rich connection between the HIP Hessian prediction in this work and other rotation-equivariant matrices of interest, like Hamiltonian prediction. For example, PhisNet Unke et al. (2021), QHNet Yu et al. (2023), HELM Kaniselman et al. (2025), and WANet Li et al. (2025) predicts the Fock matrix, and Graph2Mat Febrer et al. (2025) targets the density matrix. MōLe (Thiede et al., 2026; Burger et al., 2026) predicts coupled cluster amplitudes from Hartree–Fock orbitals.

A.3 Traditional methods for calculating Hessians

A.3.1 Analytical Hessians via CPKS

The gold standard for computing nuclear Hessians is to derive them analytically. In Kohn-Sham DFT, this is done with the Coupled Perturbation Kohn-Sham (CPKS) method. This is much more complex and computationally demanding than energy or force calculations. The difficulty arises from the need to calculate the response term of the electron density, which requires understanding how the self-consistent field (SCF) solution to the Kohn-Sham equations changes with changes in nuclear coordinates. To see why this is necessary, write the DFT energy in terms of the density matrix $\mathbf{P}(\mathbf{R}) = (\mathbf{C})^T(\mathbf{R})\mathbf{S}(\mathbf{R})\mathbf{C}(\mathbf{R})$ with molecular orbital coefficients $\mathbf{C}$ and overlap matrix $\mathbf{S}$:

$$E[\mathbf{P}] = \text{Tr}[h\mathbf{P}] + \frac{1}{2}\text{Tr}[\mathbf{J}[\mathbf{P}]] + \rho(\mathbf{P}) \quad (10)$$

To get the force, so the nuclear gradient, we first define the Lagrangian to enforce normalization of the wavefunction:

$$\mathcal{L}(C, \epsilon, \mathbf{R}) = E[\mathbf{R}] - \text{Tr}[\epsilon(\mathbf{C}^T \mathbf{S} \mathbf{C} - \mathbf{I})] \quad (11)$$

We then get the nuclear gradient by differentiating $\mathcal{L}$. The derivative w.r.t. nuclear component x is:

$$\frac{d\mathcal{L}}{dx} = \frac{\partial \mathcal{L}}{\partial x} + \underbrace{\frac{\partial \mathcal{L}}{\partial \mathbf{C}}}_{=0} \frac{\partial \mathbf{C}}{\partial x} - \text{Tr} \left[\epsilon \mathbf{C}^T \frac{\partial \mathbf{S}}{\partial x} \mathbf{C} \right] + 2\text{Tr} \left[(\mathbf{S} \mathbf{C} \epsilon)^T \frac{d\mathbf{C}}{dx} \right] \quad (12)$$

$$= \frac{\partial E}{\partial x} - 2\text{Tr} \left[(\mathbf{S} \mathbf{C} \epsilon)^T \frac{d\mathbf{C}}{dx} \right] - \text{Tr} \left[\epsilon \mathbf{C}^T \frac{\partial \mathbf{S}}{\partial x} \mathbf{C} \right] + 2\text{Tr} \left[(\mathbf{S} \mathbf{C} \epsilon)^T \frac{d\mathbf{C}}{dx} \right] \quad (13)$$

$$= \frac{\partial E}{\partial x} - \underbrace{\text{Tr} \left[\epsilon \mathbf{C}^T \frac{\partial \mathbf{S}}{\partial x} \mathbf{C} \right]}_{\text{"Pulay terms"}} \quad (14)$$

Due to the stationarity $\frac{\partial \mathcal{L}}{\partial \mathbf{C}} = 0$ we did not need to calculate the density response $\frac{\partial \mathbf{C}}{\partial x}$, which is why calculating forces is cheap, on the same order as the cost of energies. However, once we want

Hessians, we need to differentiate equation 11 twice:

$$\frac{d^2\mathcal{L}}{dx dy} = \frac{\partial^2\mathcal{L}}{\partial x \partial y} + \frac{\partial^2\mathcal{L}}{\partial \mathbf{C} \partial x} \frac{d\mathbf{C}}{dy} + \frac{\partial^2\mathcal{L}}{\partial \epsilon \partial x} \frac{d\epsilon}{dy} \quad (15)$$

$$= \frac{\partial^2\mathcal{L}}{\partial y \partial x} - \left[\frac{\partial^2\mathcal{L}}{\partial \mathbf{C} \partial x} \quad \frac{\partial^2\mathcal{L}}{\partial \epsilon \partial x} \right] \left[\frac{\partial^2\mathcal{L}}{\partial \mathbf{C}^2} \quad \frac{\partial^2\mathcal{L}}{\partial \mathbf{C} \partial \epsilon} \quad \frac{\partial^2\mathcal{L}}{\partial \epsilon \partial \mathbf{C}} \quad \frac{\partial^2\mathcal{L}}{\partial \epsilon^2} \right]^{-1} \left[\frac{\partial^2\mathcal{L}}{\partial \mathbf{C} \partial y} \quad \frac{\partial^2\mathcal{L}}{\partial \epsilon \partial y} \right]. \quad (16)$$

Solving the linear system of equations scales formally as $O(N^5)$ in computation and $O(N^4)$ in memory. Similar to energy calculations, density fitting, sparsification (integral screening), and other optimizations can reduce the cost for CPKS. Still, the scaling remains worse in both computation and memory compared to energy calculations.

A.3.2 Hessians via finite differences

A major downside of the CPKS approach to Hessians is its highly complex implementation and need for exchange-functional-specific derivations. Therefore, many new methods do not support Hessians, and most codes resort to finite-difference calculations in such cases. Using central differences of the forces along the $3N$ Cartesian directions, the Hessian elements are constructed from

$$\mathbf{H}_{ij} \approx \frac{\mathbf{F}_i(\mathbf{R} + h\mathbf{e}_j) - \mathbf{F}_i(\mathbf{R} - h\mathbf{e}_j)}{2h} \quad (17)$$

This requires $2 \times 3N$ displaced *gradient* calculations. Each gradient calculation scales similarly to the energy calculation with $O(N^4)$, leading to a $O(N^5)$, the same as naive CPKS, although typically with more numerical noise and a much larger prefactor because we have to repeatedly converge an SCF, as opposed to solving a large linear system in analytical Hessians.

A.3.3 Hessians from MLIPs

MLIPs have gained widespread popularity for their ability to accurately approximate molecular energies and forces. The forces can either be directly predicted (Gasteiger et al., 2021; Liao et al., 2024), or calculated as the derivative of the MLIP energy using automatic differentiation. Previous work has shown that the Hessian can be calculated using AD to some success (Yang et al., 2024; Yuan et al., 2024; Gönnheimer et al., 2025; Rodriguez et al., 2025; Williams et al., 2025; Cui et al., 2025). Although AD Hessians are much cheaper to calculate compared to DFT, they come with two major downsides. First, accurately modeling the energy and forces does not guarantee low Hessians errors (Yuan et al., 2024), but requires dedicated training (Zhao et al., 2025; Cui et al., 2025). Similar to how adding force labels dramatically improves autograd forces over energy-only training Christensen & Von Lilienfeld (2020), we need Hessian labels in practice for accurate autograd Hessians. Second, autograd Hessians are expensive. Since the Hessian is the derivative of the force with respect to the $3N$ atom coordinates, the cost of one autograd Hessian is $3N$ times the cost of a regular MLIP forward pass. Assuming the force calculation with an MLIP is usually $O(N)$ to $O(N^2)$, the total cost of the Hessian calculation is $O(N^2)$ to $O(N^3)$ in time and memory. Formally, to calculate the full $3N \times 3N$ Hessian, one needs $3N$ Hessian vector products (HVPs) with the unit vectors $\mathbf{v} = \mathbf{e}_i, i = 0, \dots, 3N - 1$, each yielding one column of the Hessian. While the HVPs can theoretically be parallelized, the $O(N^2)$ - $O(N^3)$ memory footprint requires even medium-sized systems to be computed sequentially. Alternatively, we can obtain Hessians using finite differences (Gönnheimer et al., 2025; Fu et al., 2025; Wander et al., 2025), which requires less memory but $6N$ forward passes. Rodriguez et al. (2025) in particular showed that including the Hessian in the loss leads to better extrapolation of energy and force predictions, less data for a given accuracy, and improved stability in MD. This comes at the cost of approximately $25\times$ longer training times, which makes AD Hessians prohibitive to train.

A.4 Hessians in Molecular Optimization

A.4.1 Geometry optimization

Geometry optimization locates minima $\text{argmin}_{\mathbf{R}} E(\mathbf{R})$ of the potential energy surface starting from non-equilibrium molecular geometries. It underpins most workflows from global searches Christiansen et al. (2022); Rønne et al. (2022) to high-throughput screening Westermayr et al. (2023); Rønne (2024). The efficiency of geometry optimization depends on four key factors: (i) the choice of coordinate

system, (ii) the quality of the Hessian approximation, (iii) the update strategy, and (iv) the step size control. Cartesian coordinates are straightforward but strongly coupled, limiting step size and slowing convergence for flexible systems. In contrast, internal coordinates are better aligned with chemically meaningful motions, separate stiff and soft modes, and allow constraints to be imposed naturally. These properties often lead to substantially fewer iterations. Some of the most widely used optimizers are the first-order FIRE (Bitzek et al., 2006) optimizer and the second-order rational function optimization (RFO) optimizer.

RFO Rational function optimization (RFO) (Simons et al., 1983; Banerjee et al., 1985) is a commonly used second-order optimization technique in molecular geometry optimization. RFO starts with a [2/2]-Padé-expansion of the energy:

$$E(\mathbf{R}_t + \Delta\mathbf{x}) - E(\mathbf{R}_t) \approx \frac{\mathbf{g}^\top \Delta\mathbf{x} + \frac{1}{2} \Delta\mathbf{x}^\top \mathbf{H} \Delta\mathbf{x}}{1 + |\Delta\mathbf{x}|^2} \quad (18)$$

The extrema $\Delta\mathbf{x}'$ of this surrogate are given by the solution of the generalized eigenvalue problem:

$$\begin{bmatrix} \mathbf{H} & \mathbf{g} \\ \mathbf{g}^\top & 0 \end{bmatrix} \begin{bmatrix} \Delta\mathbf{x}' \\ 1 \end{bmatrix} = \lambda \begin{bmatrix} I & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \Delta\mathbf{x}' \\ 1 \end{bmatrix}. \quad (19)$$

An attractive property of RFO is that it can be used both for minimizing $E(\mathbf{R})$ as well as for finding saddle points (important for transition state search, we use the restricted step RFO (RS-RFO) for minimization 19, we get an update pointing to the minimum; if we select the second eigenvector, we get a transition state update. In contrast to Newton-Raphson optimization, RFO is also robust to indefinite Hessians. For detailed derivation, see (Simons et al., 1983; Banerjee et al., 1985). In this paper, for minimization, we are using restricted step RFO (RS-RFO), which adds a trust region to prevent unphysically large steps. For the transition point search, we are using restricted step partitioned RFO (RS-P-RFO), a slight modification that treats the subspaces with negative and positive Hessian eigenvalues separately.

BFGS To apply RFO in practice, we need a Hessian at each step. As computing full Hessians is usually too expensive, a common practice is to maintain an approximate Hessian using the BFGS quasi-Newton scheme. BFGS updates approximate Hessians $\mathbf{B}_k$ of the true Hessian $\mathbf{H}(\mathbf{R}_k)$ using the update equation

$$\mathbf{B}_{k+1} = \mathbf{B}_k - \frac{\mathbf{B}_k \mathbf{s}_k \mathbf{s}_k^\top \mathbf{B}_k}{\mathbf{s}_k^\top \mathbf{B}_k \mathbf{s}_k} + \frac{\mathbf{y}_k \mathbf{y}_k^\top}{\mathbf{y}_k^\top \mathbf{s}_k}, \quad \mathbf{y}_k^\top \mathbf{s}_k > 0, \quad (20)$$

where the curvature condition $\mathbf{y}_k^\top \mathbf{s}_k > 0$ (typically ensured by a Wolfe line search) guarantees that $\mathbf{B}_{k+1}$ remains positive-definite, which is desirable for minimization. In the RFO framework, $\mathbf{B}_k$ simply replaces the exact Hessian $\mathbf{H}$ in the augmented eigenvalue problem in equation 19.

A.4.2 Transition State Search

Transition states are first-order saddle points on the PES. They represent the maximum barrier along the minimum energy path (MEP) between two minima. Identifying transition states is crucial for understanding reaction mechanisms and mapping reaction networks. A first-order saddle point is characterized by having exactly one negative eigenvalue of the Hessian. Over the years, various computational methods have been developed to describe MEPs and locate transition states.

Computational methods for transition state search can broadly be categorized as *single-ended* and *double-ended*. In double-ended methods, we know the product and reactant states (the two minima on the energy surface) and seek an interpolation along the MEP that passes through the TS. In contrast, in single-ended methods, we only know a starting point and climb the energy surface to find a nearby transition state. Often, double-ended methods are used to give good initial guesses, which are then refined by single-ended methods. In this study, we use the Growing String method to find the initial states, which we then refine using RS-P-RFO (see above).

Different double-ended methods are distinguished by how they grow the interpolations between product and reactant states. The most common approaches are the nudge elastic band method (Henkelman & Jónsson, 1999; Henkelman et al., 2000) and the growing string method (GSM) (Peters et al., 2004; Zimmerman, 2015).

The nudged elastic band method aims to find the minimum energy path between given initial and final configurations. A set of images connecting these states is linked by classical spring forces, forming an elastic band. Each image experiences a total force comprising the spring force along the local tangent and the true force perpendicular to it. The images are simultaneously optimized to trace the MEP. To converge to the transition state, a climbing image is introduced, for which the force along the band is inverted, leaving only the perpendicular component of the true force. This drives the climbing image up the potential energy surface along the band while descending perpendicularly, reaching the transition state along the MEP.

The growing string method (GSM) locates transition states by incrementally constructing a discrete path of structures on the potential energy surface. Starting from the reactant and product endpoints, two path fragments are iteratively grown together. At each iteration, a new node on the string is added along the local tangent direction and relaxed orthogonally to the path according to the force

$$\mathbf{F}_\perp = \mathbf{F} - (\hat{\mathbf{t}}^\top \mathbf{F}) \hat{\mathbf{t}}, \quad (21)$$

where $\mathbf{F} = -\nabla E$ is the force and $\hat{\mathbf{t}}$ is the local tangent unit vector. Convergence is typically accelerated by approximate Hessians, constructed in delocalized internal coordinates and updated by quasi-Newton schemes, which enable eigenvector-guided optimization while avoiding full second-derivative evaluations (Zimmerman, 2015). Once the two fragments merge, the full string is reparameterized to maintain uniform node distance, and the highest-energy node serves as a transition state estimate, which can be further refined using Hessian-based eigenvectors (Peters et al., 2004).

A.4.3 Frequency analysis for extrema classification

The final step in a transition state search or minima optimization is to confirm convergence via frequency analysis. A transition state is identified by the Hessian having exactly one negative eigenvalue (or equivalently, one imaginary frequency of the mass-weighted Hessian), whereas a minimum contains no imaginary frequencies. To carry out frequency analysis, one must first remove the five or six redundant degrees of freedom, corresponding to the invariance of the energy under rotation and translation. This is done by mass-weighting the Hessian and performing an Eckart projection (Louck & Galbraith, 1976). Then, the projected matrix is decomposed into its eigenvalues, with all positive eigenvalues indicating a minimum and exactly N negative eigenvalues indicating the presence of an order- N transition state.

A.4.4 Zero-point energy

Zero-point corrections account for the quantum mechanical vibrational energy that molecules have at absolute zero temperature. To calculate the ZPE, one relaxes a geometry to an extrema, computes the Hessian, and sums the frequencies $ZPE = \frac{\hbar}{2} \sum_i \sqrt{\tilde{\lambda}_i}$. Here, $\tilde{\lambda}_i$ are eigenvalues of the mass-weighted Hessian $\tilde{\mathbf{H}}_{AB} = \mathbf{H}_{AB}/m_A m_B$, and $\hbar$ is Planck’s reduced constant. One is usually interested in the relative ZPE between reactant and product states: $\Delta ZPE = ZPE(\mathbf{R}_R) - ZPE(\mathbf{R}_P)$. The relative ZPE is relevant for reaction thermochemistry, as ΔZPE enters as a correction to the reaction free energy ΔG and therefore the equilibrium constant, which relates forward and backward rates via detailed balance.

A.5 Hessian properties

The Hessian is a symmetric matrix $\mathbf{H} = \mathbf{H}^\top$ where each sub-block transforms under rotation as a Cartesian tensor

$$\mathbf{H}_{I,J} \xrightarrow{\mathbf{Q}} \mathbf{Q} \mathbf{H}_{I,J} \mathbf{Q}^\top \quad (22)$$

or equivalently

$$\mathbf{H} \xrightarrow{\mathbf{Q}} (\mathbf{I}_N \otimes \mathbf{Q}) \mathbf{H} (\mathbf{I}_N \otimes \mathbf{Q})^\top \quad (23)$$

The symmetry follows directly from Schwarz’s theorem, which states that for every scalar function $E(\mathbf{R})$ that has continuous partial second derivatives in the neighborhood of a point $\mathbf{R}_0$, the partial second derivatives commute

$$\left(\frac{\partial^2 E}{\partial \mathbf{R}_i \partial \mathbf{R}_j} \right)_{\mathbf{R}=\mathbf{R}_0} = \left(\frac{\partial^2 E}{\partial \mathbf{R}_j \partial \mathbf{R}_i} \right)_{\mathbf{R}=\mathbf{R}_0} \quad (24)$$

The transformation law under rotation follows straightforwardly from the chain rule: First, write the rotated coordinates as

$$\mathbf{R}' = (\mathbf{I}_N \otimes \mathbf{Q})\mathbf{R} \frac{\partial \mathbf{R}'}{\partial \mathbf{R}} = (\mathbf{I}_N \otimes \mathbf{Q}) \quad (25)$$

Then, define the energy in the rotated frame

$$E'(\mathbf{R}') = E((\mathbf{I}_N \otimes \mathbf{Q})^\top \mathbf{R}') = E(\mathbf{R}) \quad (26)$$

Then the first order derivative is

$$\nabla_{\mathbf{R}'} E'(\mathbf{R}') = (\mathbf{I}_N \otimes \mathbf{Q}) \nabla_{\mathbf{R}} E'(\mathbf{R}') \quad (27)$$

$$= (\mathbf{I}_N \otimes \mathbf{Q}) \nabla_{\mathbf{R}} E((\mathbf{I}_N \otimes \mathbf{Q})^{-1} (\mathbf{I}_N \otimes \mathbf{Q}) \mathbf{R}') \quad (28)$$

$$= (\mathbf{I}_N \otimes \mathbf{Q}) \nabla_{\mathbf{R}} E(\mathbf{R}) \quad (29)$$

showing the equivariance of the forces. The second-order derivative is

$$\mathbf{H}' = \nabla_{\mathbf{R}'}^2 E'(\mathbf{R}') = \nabla_{\mathbf{R}'} ((\mathbf{I}_N \otimes \mathbf{Q}) \nabla_{\mathbf{R}} E(\mathbf{R})) \quad (30)$$

$$= (\mathbf{I}_N \otimes \mathbf{Q}) \nabla_{\mathbf{R}'} \nabla_{\mathbf{R}} E(\mathbf{R}) \quad (31)$$

$$= (\mathbf{I}_N \otimes \mathbf{Q}) \nabla_{\mathbf{R}} \nabla_{\mathbf{R}} E(\mathbf{R}) (\mathbf{I}_N \otimes \mathbf{Q})^\top \quad (32)$$

$$= (\mathbf{I}_N \otimes \mathbf{Q}) \mathbf{H} (\mathbf{I}_N \otimes \mathbf{Q})^\top \quad (33)$$

which shows the Hessian’s equivariance.

A.5.1 Non-integrability of Hessians

Instead of relying on automatic differentiation, we propose to directly predict Hessians with an equivariant neural network. Similarly, direct force prediction has been proposed before (Gasteiger et al., 2021; Liao et al., 2024). While direct force prediction introduces non-integrable errors (non-conservative fields), these errors only cause unbounded blowups when integrated over time, as in Molecular Dynamics (MD). In contrast, for geometry optimization and standard Hessian applications (ZPE, transition state search, preconditioning), these errors remain bounded and merely limit convergence precision rather than accumulating. Therefore, direct Hessian predictions are safe for mainstream computational chemistry tasks, analogous to widely used but non-integrable methods like BFGS.

Since direct Hessian predictions are not formulated as derivatives, they are not guaranteed to be integrable. It is important to understand when a non-integrability error can lead to catastrophically wrong downstream predictions similar to direct force predictions in MD simulations. Let’s first review the more familiar case: Why do direct force predictions fail in MD simulations but succeed in geometry optimization? Suppose we directly predict forces with error $e(x) = F_{\text{pred}}(x) - F(x)$. We can decompose the error in a conservative and a non-conservative field using the Helmholtz decomposition: $e(x) = \nabla \phi(x) + S(x)$. $\phi \in \mathbb{R}$ is a scalar potential, and S is a vector field $S(x) \in \mathbb{R}^{3N}$ that is divergence-free $\nabla \cdot S(x) = 0$ but not curl-free $\exists x$ s.t. $\nabla \times S(x) \neq 0$. Now consider how the energy of the system changes over time in an MD simulation:

$$\begin{aligned} \Delta E' &= \int_0^T F_{\text{pred}}(x(t)) v(t) dt = \int_0^T (F(x(t)) + e(x(t))) v(t) dt \\ &= \int_0^T (F(x) + \nabla \phi(x) + S(x)) v(t) dt \\ &= E(T) - E(0) + \underbrace{\phi(T) - \phi(0)}_{\text{error bounded with time}} + \underbrace{\int_0^T S(x) v(t) dt}_{\text{error unbounded with time}} \end{aligned}$$

Since MD simulations use large T , non-conservative forces lead to the blowups in error.

In geometry optimization we define $E' = E + \phi$, then follow the gradient descent ODE:

$$\frac{dE'(x(t))}{dt} = \nabla E'(x(t)) \cdot \frac{dx(t)}{dt} \quad (34)$$

$$= -\nabla E'(x(t)) \cdot (\nabla E(x) + e(x)) \quad (35)$$

$$= -\nabla E'(x(t)) \cdot (\nabla E(x) + \nabla \phi(x) + S(x)) \quad (36)$$

$$= -\nabla E'(x(t)) \cdot (\nabla E'(x) + S(x)) \quad (37)$$

$$= -|\nabla E'(x(t))|^2 + \nabla E'(x) \cdot S(x) \quad (38)$$

Using the general inequality $2ab \leq a^2 + b^2$:

$$\frac{dE'(x(t))}{dt} = -|\nabla E'(x(t))|^2 + \nabla E'(x) \cdot S(x) \quad (39)$$

$$\leq -|\nabla E'(x(t))|^2 + \frac{1}{2}|\nabla E'(x(t))|^2 + \frac{1}{2}|S(x)|^2 \quad (40)$$

$$= -\frac{1}{2}|\nabla E'(x(t))|^2 + \frac{1}{2}|S(x)|^2 \quad (41)$$

Therefore, as long as $|\nabla E'(x(t))|^2 > |S(x)|^2$, the direct predicted force minimizes the energy plus some error potential similar to AD forces. Once we are close to convergence, we have $|\nabla E'(x(t))|^2 \leq |S(x)|^2$ and stop converging or enter a limit cycle. This means, we will converge to an extrema on an energy surface E' within a distance of $O(|S|)$. In practice, we are interested in how close we converge to the extrema of the DFT energy E . For this, it does not matter if the error stems from the error potential ϕ shifting the extrema, or S inducing a small limit cycle. As there is no time-dependent error blow-up, it is therefore safe to use direct forces for geometry optimization.

Now we can do the same for the directly predicted Hessian $H_{\text{pred}}(x) = H(x) + e(x)$: Denote the error again as $e(x) = \nabla^2 \phi(x) + S(x)$. As before, we would get errors scaling with integration length if we tried to calculate quantities of the form

$$Q(T) = \int_0^T f(x(t), H_{\text{pred}}(x(t))) dx(t) \quad (42)$$

leading to large errors for large T .

The key advantage is that one of the mainstream tasks integrates over a function of the Hessian for large T .

ZPE and geometry classification are single-point calculations; therefore, the errors $\nabla^2 \phi(x)$ and $S(x)$ both behave the same, one error is not "worse" than the other. Mode following during the transition state search is equivalent to the geometry-optimization analysis with direct forces above. Energy minimization with the Hessian preconditioner is also unproblematic, as the preconditioner cannot change the minimum; it only changes the convergence speed.

In fact, it is common practice to use BFGS updates for transition state search and energy minimization, and BFGS updates do not have any notion of integrability either.

A.6 Batched AD Hessian performance

To test the performance of batched Hessians, we measure the inference time per processed sample for increasingly larger batch sizes and plot the results in 6. We observe rapid degradation in the AD Hessian speed with increasing batch size. To explain this, we have to understand exactly how AD treats Hessians. From the point of view of the AD engine, there is no difference between a batch of two molecules of system size N_A and N_B , and a larger system of size $N_A + N_B$. In both cases, the MLIP function takes in an array of dimension $3N_A + 3N_B$, and returns either a single scalar $E_A + E_B$ or even worse, an array $[E_A, E_B]$, so mathematically the energy function looks like $E(\mathbf{R}) : \mathbb{R}^{3N_A+3N_B} \rightarrow \mathbb{R}$ or $E(\mathbf{R}) : \mathbb{R}^{3N_A+3N_B} \rightarrow \mathbb{R}^2$. Consequently, a Hessian of this function is a block diagonal matrix of dimension $3(N_A + N_B) \times 3(N_A + N_B)$, or even $2 \times 3(N_A + N_B) \times 3(N_A + N_B)$. Therefore, the memory costs grow quadratically with the batch size. Since we have to implement the Hessian calculation using sequential Hessian-Vector products to avoid running out of memory, we cannot even take advantage of parallelization in batched Hessian processing.

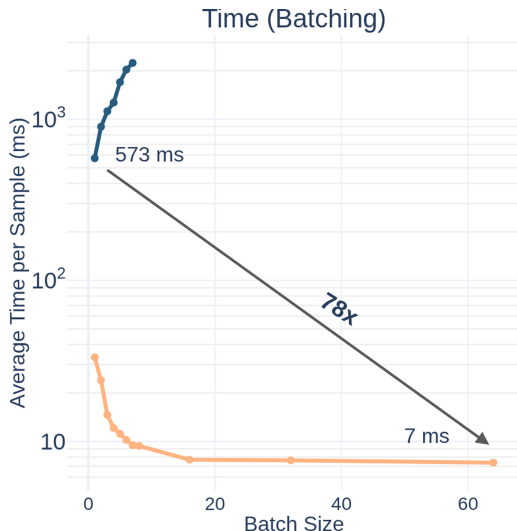


Figure 6: **Compute scaling of HIP compared to AD.** We are measuring the time per sample it takes for increasingly larger batch size. While HIP becomes faster per sample as it efficiently parallelizes, AD becomes slower.

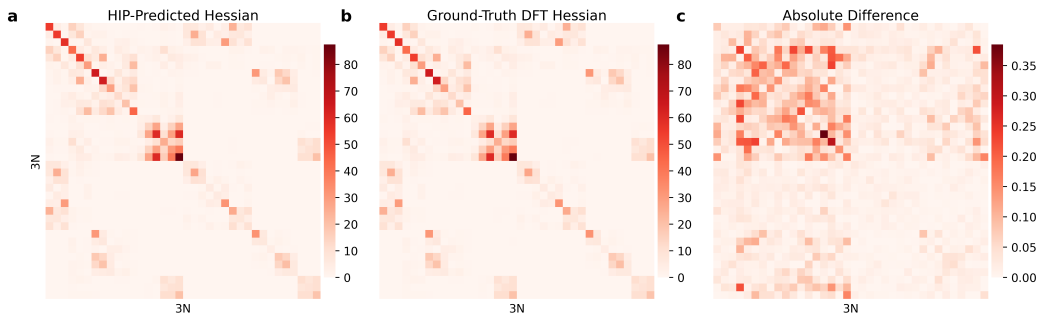


Figure 7: **HIP compared to ground-truth DFT Hessian.** The horizontal and vertical axis index the $(3N_{Atoms})^2$ entries of the Hessian. We depict the absolute values of test sample C5H6O in $\text{eV}/\text{\AA}^2$.

A.7 Additional results

A.7.1 Loss function

We are trying to design a loss function that emphasizes the lowest lying eigenvalues/eigenvectors. To design such a loss function, one could naively compute a loss directly as $L = \sum_i^k |\mathbf{v}_i^{\text{predict}} - \mathbf{v}_i^{\text{true}}|$. This requires computing the eigenvalues and eigenvectors of the true and predicted Hessian via an eigenspectrum decomposition. Unfortunately, backpropagation through such an eigenspectrum decomposition is numerically unstable (van der Aa et al., 2007). First, gradients computed using the eigenvectors tensor will only be finite when $\mathbf{H}$ has distinct eigenvalues. Furthermore, if the distance between any two eigenvalues is close to zero, the gradient will be highly sensitive, as it depends on the eigenvalues λ_i through the computation of $\min_{i \neq j} \frac{1}{\lambda_i - \lambda_j}$. Instead, we propose the following subspace loss.

In the following section, we compare the eigenspace loss that we introduced in equation 9 to the standard choice of mean square error (MSE) and mean absolute error (MAE).

In particular, we emphasize the subspace $k = 8$. This is because transition state search, ZPE, and frequency analysis for extrema classification all depend on the two lowest-lying eigenvectors and eigenvalues. We choose $k = 8$ instead of $k = 2$ to account for the usual 6 redundant degrees of freedom, which we do not remove during training for numerical stability.

Loss	Hessian ↓ eV/Å ²	Eigenvalues ↓ eV/Å ²	CosSim v_1 ↑ unitless	λ_1 ↓ eV/Å ²	λ_2 ↓ eV/Å ²	λ_1^E ↓ eV/Å ²	λ_2^E ↓ eV/Å ²	CosSim v_1^E ↑ unitless	CosSim v_2^E ↑ unitless
MSE	0.033	0.072	0.728	0.168	0.088	0.037	0.013	0.959	0.928
MAE	0.026	0.057	0.825	0.127	0.052	0.031	0.012	0.976	0.952
MAE+Sub	0.030	0.063	0.870	0.130	0.030	0.031	0.010	0.980	0.957

Table 2: Accuracy of HIP-EquiformerV2 using different loss functions. Superscript E denotes after removing redundant degrees of freedom using Eckart-projection on the mass-weighted matrix.

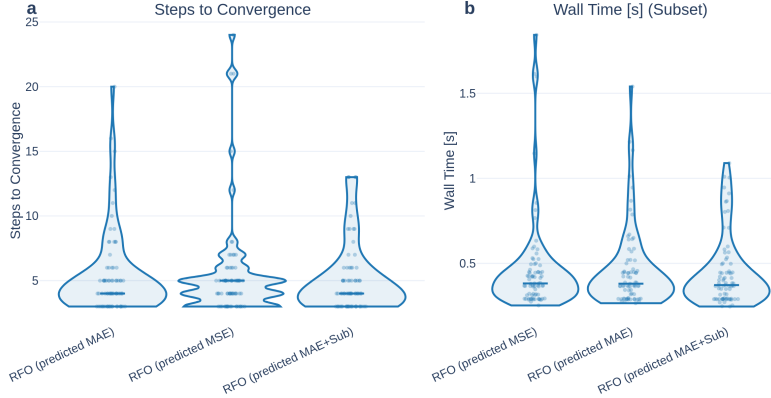


Figure 8: Geometry relaxation with HIP-EquiformerV2 trained using different loss functions

We combine the subspace loss with MAE, so the total loss becomes

$$\mathcal{L}_{\text{MAE+Sub}} = \mathcal{L}_{\text{MAE}} + \mathcal{L}_{\text{sub}}^{(k=8)} \quad (43)$$

$$= \sum_{i,j} |\mathbf{H}_{i,j} - \mathbf{H}_{i,j}^{\text{pred}}| + \sum_{i,j} \left| \mathbf{V}_{[:,k]}^{\text{T}} \mathbf{H}^{\text{pred}} \mathbf{V}_{[:,8]} - \mathbf{\Lambda}_{[:,8]} \right|_{i,j} \quad (44)$$

For simplicity, we weight both the MAE and subspace loss equally. Future work could benefit from carefully tuning the relative weights between the loss terms.

On downstream tasks, subspace loss improves over MAE and MSE in (i) transition state search, as shown in Table 4, and (ii) transition state identification frequency analysis, as shown in Table 5. We observe equal performance between MAE and the subspace loss for (i) geometry relaxation Table 8 and (ii) computing the zero-point energy Table 3.

Table 2 shows that the subspace loss improves the first eigenvector v_1 cosine similarity, and the second eigenvalue λ_2 MAE. This is consistent with the better performance of the subspace loss on transition state-related tasks, since the transition state is characterized via the two smallest eigenvalues λ_1, λ_2 , and is found by following the first eigenvector v_1 .

MSE generally underperforms compared to MAE and MAE with subspace loss. MSE has the theoretical advantage of being rotation-invariant, whereas MAE is not. In practice, however, we find that the MSE leads to a high variance (spikes) in the training loss and gradient norm across batches. We speculate that MAE outperforms MSE due to training stability.

Hessian prediction with any loss function tested significantly outperforms prior AD Hessians.

Hessian	Model	ZPE MAE (Std) [eV]	Δ ZPE MAE (Std) [eV]
MSE	HIP-EquiformerV2	0.0008 (0.0012)	0.0019 (0.0025)
MAE	HIP-EquiformerV2	0.0004 (0.0004)	0.0013 (0.0019)
MAE+Sub	HIP-EquiformerV2	0.0004 (0.0003)	0.0016 (0.0018)

Table 3: Zero-point energy by HIP-EquiformerV2 for different loss functions

Loss	Model	TS Success	RFO Converged	Both
MSE	HIP-EquiformerV2	718	669	659
MAE	HIP-EquiformerV2	740	693	683
MAE+Sub	HIP-EquiformerV2	750	705	698

Table 4: Transition state search with HIP-EquiformerV2 trained with different loss functions

Name	TPR $\uparrow$	FPR $\downarrow$	FNR $\downarrow$	TNR $\uparrow$	Precision $\uparrow$	Accuracy $\uparrow$	Accuracy All $\uparrow$
MSE	88%	11%	12%	89%	86%	89%	86%
MAE	93%	7%	7%	93%	91%	93%	91%
MAE+Sub	93%	6%	7%	94%	92%	94%	92%

Table 5: Identifying transition states (left) and any extrema (rightmost column) via frequency analysis using HIP-EquiformerV2 for different loss functions.

A.8 Experimental Details

A.8.1 Training

The hyperparameters of our HIP-EquiformerV2 model are listed in 6. We inherit the model settings and large parts of the optimizer settings from the HORM codebase (Cui et al., 2025).

Model	Type: EquiformerV2 Layers: 4 Sphere channels: 128 Attention hidden channels: 64 Attention heads: 4 Attention alpha channels: 64 Attention value channels: 64 FFN hidden channels: 128 Activation: SiLU Distance basis: 512 (Gaussian) Radius cutoff: 12 Angstrom Cutoff Hessian: 12 Angstrom Hessian layers: 3 (1 for end-to-end) Spherical harmonics: Lmax=4, Mmax=2 Grid resolution: 18, Sphere samples: 128 Dropout: $\alpha=0$, drop path=0
Loss	MAE weight: 1 Eigen subspace k : 8 Eigen loss weight α : 1.0
Optimizer	AdamW Betas: (0.9, 0.999) AMSGrad: True Weight decay: 0 Batch size: 128 Gradient clipping: 0.1 (norm)
Learning Rate	Learning rate: 0.0005 Scheduler: StepLR Step size: 10 Gamma: 0.85
Trainer	Steps: 500k (3-5 days on a H100 GPU)

 Table 6: **Hyperparameters.** For simplicity, we use the same hyperparameters for HIP as HORM (Cui et al., 2025).

Training AD Hessians is exceptionally more expensive than training HIP. The memory cost of computing loss gradients through the full $3N \times 3N$ Hessian is prohibitive. Instead, the baseline models from Cui et al. (2025) randomly sample one or two columns of the Hessian via random Hessian-vector products at each training step. Despite this, training AD Hessians remains exceedingly expensive. The HORM paper states that they use 2 HVPs at a batch size of 128. Their checkpoint metadata shows a per-device batch size of 8, suggesting a $16 \times$ A30 GPU setup for 1 million gradient steps. In contrast, we trained HIP on a single H100 GPU for 500k steps. Additionally, each HIP training step costs 1/5 as much as AD Equiformer training on our hardware. Finally, we observe that

HIP fully converges after 300-500k training steps, whereas AD Hessians require multiple steps due to the limited supervision and the inability to use the full Hessian.

Model	Hessian Columns	Batch Size	BZ / GPU	GPU	# GPUs	Training Steps
AlphaNet	1	32	16	H20	2	688,000
LeftNet-CF	1	64	16	A30	8	192,000
LeftNet-DF	2	64	16	A30	8	120,000
EquiformerV2 (E-F)	0	128	8	A30	16	384,000
EquiformerV2	2	128	8	A30	16	1,008,000
HIP-EquiformerV2	3N (all)	128	128	H100	16	500,000

Table 7: **Summary of training resources.** Comparing HIP to checkpoints from HORM (Cui et al., 2025).

A.8.2 Geometry Optimization

We compare our RFO-based optimization (Besalú & Bofill, 1998) using predicted Hessians (RFO predicted), and our RFO with BFGS updates with initial predicted Hessian (RFO-BFGS predicted init) against first-order methods, quasi-second-order BFGS variants, and exact second-order approaches obtained via automatic differentiation or finite differences.

Our baselines include steepest descent with backtracking line search (SteepestDescent) and the first-order method FIRE (Bitzek et al., 2006). For quasi-second-order methods, we consider RFO with BFGS updates initialized from the identity (RFO-BFGS). For full second-order methods, we include RFO with either automatic differentiation or finite difference Hessians. We run relaxations on 80 reactant geometries from the Transition1x validation set (which is different from HORM-T1x), that we noise with 0.5 Å RMS. We use the RS-RFO and redundant coordinate implementation in pysisyphus (Steinmetzer et al., 2021; Herbol et al., 2017). We set the convergence criteria to "Gaussian default" (see A.8.5), with a budget of 150 steps.

A.8.3 Transition State Search with ReactBench

We use the recently introduced ReactBench benchmark (Zhao et al., 2025), which relies on pysisyphus (Steinmetzer et al., 2021) and PyGSMALdaz et al. (2025) for implementation of the optimization routines. The workflow consists of generating an initial guess using the growing string method (Aldaz et al., 2025; Zimmerman, 2015), followed by local search using the restricted-step partitioned rational-function-optimization method (RS-P-RFO) (Besalú & Bofill, 1998). Finally, convergence to the correct transition state is confirmed by frequency analysis and following the intrinsic reaction coordinate (IRC). Following previous work (Zhao et al., 2025; Cui et al., 2025), we report success metrics of each step:

1. GSM successfully converged below a force RMS of $5e^{-5}$ Hartree/Bohr within 100 iterations ("GSM Success")
2. After RS-P-RFO, the frequency analysis determines geometry as a true transition state ("TS Success")
3. IRC converged to the same criteria and yields the original initial reactant and product ("IRC Verified").

We then treat the samples successfully passing (a)-(c) as transition state proposals, and verify for a random subset of 100, if the DFT Hessians have one negative eigenvalue and the DFT force RMS is below $2e^{-3}$ Ha/Bohr. We additionally report if RS-P-RFO converged to "Gaussian default" (A.8.5) within 50 steps.

A.8.4 Zero-Point Energy

Starting from 80 reactant and product geometries from the Transition1x validation set, we relax the geometries using DFT and RS-RFO (BFGS, identity initialization) until "Gaussian tight" convergence (see A.8.5). For a given method m , the reaction ZPE change is

$$\Delta\text{ZPE}_m = \text{ZPE}_m(\text{product}) - \text{ZPE}_m(\text{reactant}).$$

Setting	Max Force	RMS Force	Max Step	RMS Step	Used for
Gaussian default	4.5e-4	3.0e-4	1.8e-3	1.2e-3	Relaxation, TS search 4.5.1
Gaussian tight	1.5e-5	1.0e-5	6.0e-5	4.0e-5	ZPE A.8.4

Table 8: Convergence criteria used in our experiments

The $\Delta\Delta$ DFT error for a model is

$$\Delta\Delta\text{ZPE} = |\Delta\text{ZPE}_{\text{model}} - \Delta\text{ZPE}_{\text{DFT}}|.$$

We use the same level of theory for model relaxation as for model training. Then we compute the ZPE from the Eckart-projected, mass-weighted Hessian, as predicted by the different models, and compare it to DFT. We report all energies in eV per molecule.

A.8.5 Convergence Criteria

Throughout our experiments, we adopt the following convergence criteria, as shown in table 8, set forth by the Gaussian software and widely used across different codebases. All criteria need to be met for a geometry to be considered converged.

A.8.6 Glycine proton transfer

To assess whether the predicted Hessians encode physically meaningful curvature near a transition state, we selected a reaction for which the relevant chemical motion can be described by simple internal coordinates. We first annotated all non-training reactions in Transition1x by comparing reactant and product bond graphs, excluding the training split and considering both the validation and test splits. For each reaction, we recorded the bonds formed and broken, the number of connected molecular components, and a coarse reaction-family label. We then prioritized reactions with a small number of bond changes and an unambiguous two-dimensional representation of the reaction coordinate.

We chose the intramolecular proton transfer of glycine, present in the Transition1x test split (sample ID 5, rxn1961). Glycine is the simplest amino acid, and its neutral-zwitterionic tautomerization, $\text{NH}_2\text{CH}_2\text{COOH} \rightleftharpoons \text{NH}_3^+\text{CH}_2\text{COO}^-$, has been widely studied as a model proton-transfer process. In this reaction, the acidic carboxyl proton is transferred between an oxygen and the amine nitrogen, interconverting O–H- and N–H-bonded forms. The transferring proton is therefore naturally described by the two distances $q_{\text{NH}} = d(\text{N4}, \text{H9})$, $q_{\text{OH}} = d(\text{O3}, \text{H9})$. Previous studies of glycine tautomerization have commonly used a proton-transfer coordinate based on these distances $\xi = q_{\text{NH}} - q_{\text{OH}}$ (see e.g. Zhang et al. (2024)).

We first investigate a two-dimensional surface around the proton transfer. For each point on the $(q_{\text{NH}}, q_{\text{OH}})$ surface, we constructed an initial geometry by starting from the Transition1x (Schreiner et al., 2022) transition-state structure and placing the transferring proton to satisfy the target values of q_{NH} and q_{OH} . We excluded any pair $(q_{\text{NH}}, q_{\text{OH}})$ that violated the triangle inequality $q_{\text{NH}} + q_{\text{OH}} < d(\text{N4}, \text{O3})$ or $|q_{\text{NH}} - q_{\text{OH}}| > d(\text{N4}, \text{O3})$. The remaining structures were relaxed while constraining q_{NH} and q_{OH} , allowing all other degrees of freedom to relax. They were first relaxed with GFN2-xTB, using TBLite and ASE BFGS. They were then reoptimized at the DFT level using ORCA 6.1.1 with ! wB97X-D3 6-31G(d) TightSCF Opt. The remaining structures were first relaxed with fixed q_{NH} and q_{OH} , allowing all other degrees of freedom to relax, and were then reoptimized at the DFT level with the same two bond-distance constraints. This produced 579 DFT-relaxed geometries.

For each DFT-relaxed geometry, we computed energies, forces, and Hessians using ORCA 6.1.1 Neese (2012) with the ω B97X-D3 functional and the 6-31G(d) basis set. Geometry optimizations used ! wB97X-D3 6-31G(d) TightSCF Opt with the two proton-transfer bond distances constrained. Energies and forces were then evaluated with ! wB97X-D3 6-31G(d) TightSCF EnGrad, and Hessians were evaluated with ! wB97X-D3 6-31G(d) TightSCF Freq. The data is available at huggingface.co/andreasburger/hip/tree/main/orca_wb97x_d3_631gd_glycine_pt_dft_relaxed_579. The resulting energy surface is shown in figure 9.

We also study the one-dimensional minimum energy path for the same glycine proton-transfer reaction to evaluate model behaviour along a chemically meaningful trajectory. We used the stored

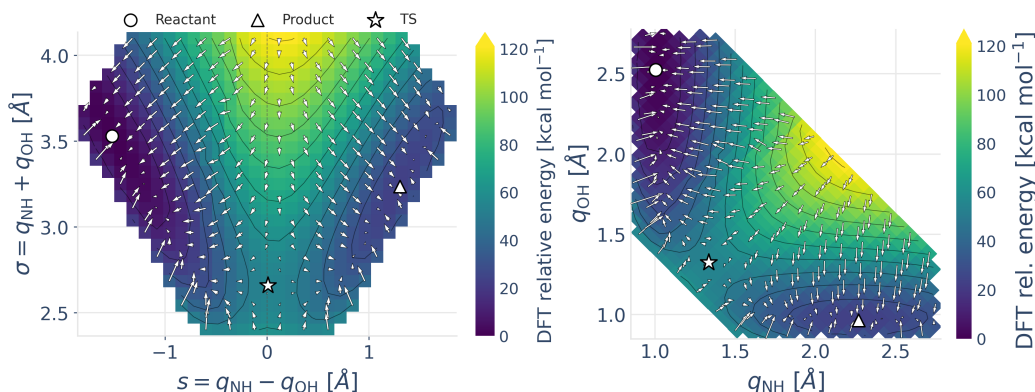


Figure 9: **Glycine**. 2D collective variable energy surface at ω B97X/6-31G(d), with forces shown as arrows. The DFT-optimized reactant and product contain no imaginary frequencies, while the DFT transition-state optimization yields one imaginary frequency, confirming the expected minimum/first-order-saddle classifications.

Transition1x record, which contains 266 geometries, which we interpreted as an initial 10-image NEB path followed by 32 saved blocks of 8 internal NEB/CINEB images. We used the final saved block of 8 with the reactant and product geometries as the 10-image minimum energy path.

To obtain a smooth trajectory for evaluation and visualization, we connected adjacent images along this path using geodesic interpolation in redundant internal-coordinate space. The original NEB images were kept fixed. This produced 150 path geometries. All methods were evaluated on the same fixed structures. Again, single-point energies, forces, and Cartesian Hessians were computed for each geometry using ORCA 6.1.1 with `! wB97X-D3 6-31G(d) TightSCF EnGrad Freq` and are available at huggingface.co/andreasburger/hip/tree/main/orca_wb97x_631gd_glycine_mep_145.

A.8.7 Size generalization on PubChem

To test Hessian prediction beyond the HORM training distribution, we assembled a benchmark of 710 neutral organic molecules containing only C, H, N, and O, with 30–100 atoms (71 atom-count bins, 10 molecules per bin). Geometries were drawn from the PubChem database via the PUG REST API. For each target atom count N , we enumerated stoichiometrically plausible neutral CHNO formulas and queried PubChem for matching compounds (‘fastformula’ endpoint, up to 12 CIDs per formula). Candidates were retained only if they satisfied (a) formal charge 0 (PubChem ‘Charge’ property) (b) composition restricted to $\{Z = 1, 6, 7, 8\}$ (c) exactly N atoms in the 3D structure.

We first attempted to retrieve PubChem 3D conformers (SDF). When no suitable 3D structure was available, we generated a conformer from the isomeric SMILES using RDKit (ETKDGv3 embedding followed by MMFF or UFF relaxation). Uncovered atom-count bins were additionally filled from deterministic RDKit-generated CHNO conformers. Reference energies, Cartesian gradients, and Hessians were computed with ORCA 6 at ω B97X/6-31G(d), TightSCF, on the fixed input geometries (charge 0, multiplicity 1). We used two single-point jobs per molecule: (i) a Freq calculation for the analytical Hessian, and (ii) an EnGrad calculation for the energy and gradient.

```
! wB97X 6-31G(d) TightSCF Freq
! wB97X 6-31G(d) TightSCF EnGrad
%pal nprocs 8 end
%maxcore 4000
```

The reference dataset and geometries are released under huggingface.co/andreasburger/hip/tree/main/orca_wb97x_631gd_chno_30_100.

For aggregate PubChem plots, we remove pathological molecules before bin-wise averaging. Outliers are identified from the per-molecule Cartesian Hessian MAE, $\text{MAE}(\mathbf{H}) = \frac{1}{9N^2} \sum_{i,j=1}^{3N} |H_{ij}^{\text{pred}} - H_{ij}^{\text{ref}}|$, where N is the number of atoms. For each atom-count bin N , let

$\{x_i\}_{i=1}^{n_N}$ denote the Hessian MAEs of molecules in that bin, with $n_N = 10$. We compute the bin median $\tilde{x}_N$ and the median absolute deviation $\text{MAD}_N = \text{median}_i(|x_i - \tilde{x}_N|)$. The modified z -score of sample i is $M_i = 0.6745 \frac{x_i - \tilde{x}_N}{\text{MAD}_N}$, following Iglewicz and Hoaglin (Iglewicz & Hoaglin, 1993). This modified z -score uses the median instead of the mean, and MAD instead of the variance, to improve robustness. The prefactor 0.6745 is the median absolute deviation of a standard normal variable. Sample i is flagged as an outlier if it exceeds $|M_i| > 10$. Outliers are detected independently for each method, and the union of flagged samples is removed from all methods so that curves remain comparable. This removes 10 of 710 PubChem molecules.

Because raw Cartesian Hessian eigenvalues mix internal and external (translation/rotation) modes, we additionally project both reference and predicted Hessians onto the vibrational subspace using Eckart projection (mass-weighted, translation-rotation removal) and compare the resulting eigenvalue spectra. The eigenvalue MAE is computed as $\text{MAE}_\lambda = \langle |\lambda_k^{\text{pred}} - \lambda_k^{\text{ref}}| \rangle$, where $\{\lambda_k\}$ are the projected vibrational eigenvalues eV .